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ABSTRACT

We decode China's industrial policies from 2000 to 2022 by employing large language models (LLMs) to extract and analyze rich information from a comprehensive dataset of 3.3 million documents issued by central, provincial, and city-level governments. Through careful prompt engineering, multistage extraction and refinement, and rigorous verification, we use LLMs to classify the industrial policy documents and extract structured information on targeted industries, policy tones (supportive or regulatory/suppressive), implementation tools, etc. Combining these newly constructed industrial policy data with micro-level firm data, we explore the political economy of industrial policy, focusing on the economic and political foundations of sector choice and its evolution over time. We find 2013 as a turning point for policy recentralization, both in the central policies' coverage and strength and in the local governments' compliance. We document spatial inefficiencies and industry-wide overcapacity as potential downsides of industrial policies.

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1 Introduction

Industrial policy is experiencing a powerful global resurgence. From the CHIPS Act aimed at bolstering semiconductor production in the U.S., to the prioritization of renewable energy by the European Green Deal and India's "Atmanirbhar Bharat" agenda for self-reliance, governments around the world are launching initiatives to secure advantages in critical sectors. China's industrial policy is widely viewed as central to its structural transformation and its rise up the global value chains. This surge of interest in industrial policy reflects a wider re-evaluation of the state's role in driving growth and innovation in an interconnected but increasingly fragmented global economy.

Academic interest in industrial policies has been revived accordingly. The early contributions were skeptical, highlighting rent-seeking and inefficiency from government intervention (Krueger, 1990; Pack, 2000). More recent work leveraging narrower scopes and stronger identification reports more favorable effects (Lane, 2025; Manelici and Pantea, 2021; Choi and Levchenko, 2025). This shift refocuses attention from a binary "works or not" debate toward design and implementation margins: which sectors are targeted, with which instruments, how the targeting diffuses across jurisdictions, and how political incentives shape these choices and their consequences. Echoing Rodrik (2004, 2009), the relevant question is not whether to practice industrial policy but *how*. Progress on these questions has been constrained by the lack of comprehensive, systematically coded policy data—especially the internal dynamics and regional variation within a single country. We fill this gap through a detailed, large-scale analysis of industrial policy documents issued by central, provincial, and city-level governments in China.

China is an especially important laboratory for studying industrial policies. First, its extensive use of industrial policies to shape sectoral development makes it pivotal for understanding the role of policy in both the successes and pitfalls of its economic development (Naughton, 2021; Naughton, Xiao, and Xu, 2023). Second, China's governance structure—central strategic guidance combined with local discretion in adoption and implementation—generates rich regional variations in policy content and tool choices. The heterogeneity across cities in their economic fundamentals, administrative and fiscal capacity, and local politicians' career incentives enables the analysis of the determinants and consequences of policy. Crucially, a unified political-legal framework within a single country reduces the potential confounders that plague cross-country studies and sharpens the inference on the underlying mechanisms. Third, the discretion afforded to local governments exposes the fundamental tension between local competition and national coordination. As local officials balance local economic fundamentals and capacities, as well as their career incentives, with upper-level guidance, their choices may not necessarily align with aggregate efficiency. The marked shift toward recentralization in recent years further provides an ideal setting to examine these coordination challenges and to assess whether centralization mitigates or exacerbates them—a central issue in the design of industrial policies.

We construct a comprehensive dataset on industrial policies by applying Large Language Models (LLMs) to a corpus of more than 3 million policy documents issued between 2000 and 2022 by all levels of government, from central to cities, in China. The LLMs first identify industrial policies and then parse these unstructured texts to extract a rich set of structured attributes: issuing government and issuing date, targeted industries, policy objectives, tone (supportive versus regulatory/suppressive), policy tools, measures of policy strength, and conditionality. This granular approach goes beyond a binary policy vs. no-policy indicator, allowing us to map the formulation, diffusion, and implementation of policy across administrative levels. As the effects of industrial policy hinge on implementation details, our granular data fill a critical gap. By linking these detailed industrial policy data to micro-level firm datasets, we can analyze how complex policy bundles shape firm entry and performance. To our knowledge, ours is the first comprehensive and granular dataset of China's industrial policies.

Deploying LLMs in this setting is challenging because long documents and complex prompts raise the risk of hallucination. Therefore, we develop a set of *hallucination-robust* LLM strategies. We decompose complex queries into sub-questions that guide step-by-step reading; provide precise definitions and counterexamples to reduce ambiguity; require responses to include the relevant source text, reasoning, and confidence score; separate text-extraction from classification to curb unwarranted inferences; and verify validity by checking that extracted passages appear in the documents post-query. Building on the LLM-agent framework (Madaan et al., 2023; Shinn et al., 2023; Li et al., 2024), we integrate responses from *multiple* LLMs and apply an iterative refinement approach for the core tasks. We also have human experts in the loop to enhance transparency and conceptual accuracy. We validate the LLM output in two ways: first, in an internal validation with 2,000 documents labeled by research assistants, the LLM-based classifier achieves high agreement with human labels and compares favorably to standard text-classification benchmarks; second, in an external validation, the resulting exposure measures predict subsequent firm entry and financial outcomes in ways consistent with policy content.

The procedure identifies roughly 0.6 million industrial policy documents. Their sheer volume underscores the importance of industrial policies in China. The number of policies increases steadily over time, with a growing share with supportive tones. Spatially, more developed regions issue more industrial policies. Across sectors, manufacturing accounts for 35% of all policies. Over time, the overall share of industrial policies targeting manufacturing has decreased slightly; however, within manufacturing, we find that the share targeting high-skill segments has increased significantly, consistent with technological upgrading. Over time, the focus of industrial policies on the agriculture sector has been declining, while the focus on the service sector has been increasing.

We classify policy tools into 20 categories. Fiscal subsidy is the most common, appearing in 49% of policies, yet more than half of policies use no subsidy—highlighting the potential bias in measures that proxy "industrial policy" with subsidy flows alone. Policy tools

appear in *bundles* and vary systematically across regions and sectors, thus cautioning against studying any single instrument in isolation. Policy tools matter for outcomes, and they are differentially effective for different purposes: fiscal subsidies, infrastructure investment, preferential land supply, and industrial clusters are most effective in stimulating new firm entry; the R&D support contributes the most to the growth of total factor productivity (TFP), followed by industrial clusters, fiscal subsidies, and consumer subsidies.

Interestingly, the choice of policy tools is not static; rather, it evolves over an industry's development phase. When a city first targets an industry and the primary challenge is to attract initial investment and secure demand for new entrants, governments tend to deploy entry- and demand-tools (e.g. fiscal subsidies, preferential land supply, tax incentives, and government procurement); however, as the targeted industries mature and the challenge shifts toward enhancing productivity and market expansion, the tool mix correspondingly shifts toward upgrading, expansion, and coordination (e.g. R&D and technology adoption, labor policy, environmental policy, equity support, consumer subsidy). These patterns suggest a certain level of sophistication in policy-making to tailor tools to the life cycle of the targeted industry.

Taken together, these descriptive patterns map the landscape of China's industrial policy. We then turn to the political economy of sectoral targeting over the past two decades. We first document how the local choice of targeted industries reflects both economic fundamentals and top-down political incentives, and we show that policy-making became markedly more centralized after 2013. We then examine the mechanisms behind this shift and its consequences.

We begin by examining the economic and political determinants of the target sectors by local governments. Economically, cities are more likely to support industries in which they have a pre-existing revealed comparative advantage (RCA) or a large local presence. They also disproportionately target sectors that are central in production networks, and they often coordinate targeting across upstream and downstream industries. Politics also matters. City-level portfolios of targeted industries exhibit substantial pass-through from upper-level priorities, and the strength of this pass-through varies systematically: it is weaker in richer cities, stronger where intra-province promotion competition is more intense, and attenuated when local leaders have personal ties to upper-level officials.

To gauge how the balance between economics and politics evolves, we compare changes over time in top-down political alignment and in alignment with the local industrial structure. We find a clear break around 2013: the cities' industrial policy portfolios become more closely aligned with their province's portfolio, while their fit with the local RCA declines. Put differently, city policies converge to provincial priorities but diverge from local comparative advantage. After 2018, the two measures partially rebalance–political alignment remains elevated but stabilizes, while the fit with local RCA recovers modestly.

We then examine the mechanisms behind this recentralization from both the central and local perspectives. At the central level, we find that its policy directives became stronger

and broader after 2013: text-based measures of policy concreteness and strength rose by roughly ten points on a scale 0-100, while the sectoral Herfindahl index of national policies declines, indicating a more dispersed portfolio of targeted sectors. The post-2013 shift is not driven by top-down mandates alone; it also operates through altered local fiscal and political incentives. As the central agenda expands and becomes more concrete, local compliance with the central directives becomes more resource-intensive. Consistent with a fiscal channel, top-down political alignment is stronger in cities with weaker own-source fiscal capacity and greater reliance on intergovernmental transfers, and the pass-through from provincial priorities to city targeting is amplified most when provincial policies deploy fiscal and financial instruments. A complementary political channel works through promotion incentives: before 2013, promotions are more strongly associated with the fit of local industrial policies to local RCA, whereas after 2013, alignment with upper-level priorities becomes more predictive—especially among leaders without personal ties to upper-level officials.

Turning to firm outcomes, we distinguish between economic alignment—targeting sectors that fit local comparative advantage—and political alignment—targeting sectors emphasized by upper-level governments. Both are associated with higher entry in targeted city–industry cells, but their performance implications differ. When local industrial policies are aligned with local RCA, the positive association between policy targeting and firm outcomes is stronger: firms exhibit higher TFP, wages, and profitability; product market markups are stable or slightly higher; and labor market wage markdowns are lower. In contrast, under political alignment, the corresponding performance gains of firms are weaker and can turn negative: TFP, wages, and profitability fall relative to economically aligned targeting, product market markups compress, and labor market wage markdowns rise. Taken together, these patterns are consistent with a mechanism in which top-down political incentives induce cities to replicate upper-level governments' portfolios of industrial policies even when sector choices are a poor match for local industrial structure, suggesting a potential efficiency cost of greater centralization.

When sectoral priorities diffuse through the policy hierarchy, many cities end up targeting the same industries at similar times. We find that this diffusion is associated with stronger entry responses in follower cities, but weaker—and in some cases negative—associations with firm performance (TFP, profits, and wages, and hence labor market wage markdowns) as adoption becomes more widespread and markets become more crowded. Implementation also becomes more "templated" over the diffusion sequence: within an industry-year, the cosine similarity between a city's policy tool vector and its provincial tool vector increases monotonically with adoption order. Despite these diminishing (and sometimes negative) returns, we find little evidence of local self-correction; conditional on having targeted an industry, continuation of the policy is more likely when entry is higher, product market markups are lower, and labor market wage markdowns are higher–precisely when a course correction would have been necessary; alignment with upper-level directives further reinforces persistence.

These micro patterns aggregate into broader changes in policy and economic structure. As central guidance expands and strengthens and top-down political alignment increases, the city-level concentration of policy-targeted industries falls, and the cross-city policy similarity rises. This combination results in *more dispersed* industrial structures *within a city* and *more similar* structures *across cities*, reducing intercity complementarities and, correspondingly, cross-city trade. Using firm-to-firm value-added tax (VAT) transactions data, we corroborate these patterns: at the city level, the intra-city share of total trade is positively correlated with a city's average policy similarity within the province; at the city-pair level, the bilateral share of trade between cities is negatively correlated with the pairwise policy similarity of the cities. Taken together, the evidence is consistent with broader, vertically aligned agendas that contribute to homogenization across cities and a greater tendency to overcapacity and local protectionism.

Related Literature. This paper contributes to the growing literature on industrial policy (see Juhász, Lane, and Rodrik, 2024 for a comprehensive review). There is a large theoretical literature on industrial policy (e.g. Baldwin, 1969; Krueger, 1990; Krugman, 1992; Harrison and Rodríguez-Clare, 2010; Lin, Monga, and Stiglitz, 2013; Itskhoki and Moll, 2019). The earlier empirical literature on industrial policy focuses mainly on describing what happens to policy-targeted industries, most of which find that industrial policies have generally been ineffective or counterproductive (Baldwin and Krugman, 1986; Head, 1994; Luzio and Greenstein, 1995; Irwin, 2000; Hansen, Jensen, and Madsen, 2003). More recent literature zooms into specific settings with a careful identification design to assess whether industrial policy elicited the desired responses from firms in the target sectors (e.g. Lane, 2025, 2020; Juhász, 2018; Choi and Levchenko, 2025). These studies produce results that are much more favorable to industrial policies.

Our contribution is threefold. First, we build a comprehensive and granular database on China's industrial policies from a large corpus of government documents, using LLM-based extraction to recover targeted sectors, policy instruments and related implementation features at fine geographic and temporal resolution. This complements existing approaches that rely on a small set of structured plans or on ex post subsidy flows, and it allows us to characterize how targeting and policy tools vary across places, over time, and across levels of government, and to examine the mechanism behind the effectiveness and pitfalls of industrial policies.

Second, the resulting data allow us to study industrial policy as a bundle of choices—sector selection, mix of policy tools, and diffusion across jurisdictions—rather than as a single intervention. We use the data to document systematic patterns in policy diffusion and in mapping from policy targeting to firm outcomes, and to analyze how these patterns change around the post-2013 shift toward greater recentralization in China's policy-making.

Third, we contribute to the literature on China's industrial policy. Existing work provides institutional background (Naughton, 2021), estimates spending magnitudes (DiPippo

et al., 2022), and evaluates particular instruments or episodes using firm-level data or structural models (Aghion et al., 2015; Barwick, Kalouptsidi, and Zahur, 2025; Branstetter, Li, and Ren, 2023).[1] Related work also highlights the role of production networks in policy design (Liu, 2019). Our research offers by far the most comprehensive description of China's industrial policy landscape, spanning over 20 years and granular at the city level, as previous research has generally been limited to national or provincial analyses relying on specific policy documents, e.g., the Five-Year Plans. We provide new insights into the political economy of industrial policies by documenting the interactions between central, provincial, and local governments in the formation and implementation of policies. Our evidence sheds light on how China's politically centralized and economically decentralized system shapes policy transmission and adaptation (Xu, 2011), and on potential inefficiencies, such as homogenization of the industrial structure, local protectionism, and overcapacity, associated with intensified vertically aligned top-down diffusion.

Methodologically, our paper relates to the rapidly growing "text-as-data" literature in economics and finance (Gentzkow, Kelly, and Taddy, 2019; Goldstein, Spatt, and Ye, 2021) and, more specifically, to recent work that uses LLMs for measurement and prediction tasks in economic and financial settings (e.g. Li et al., 2025; Lopez-Lira and Tang, 2023; Li, Tu, and Zhou, 2024; Kim, Muhn, and Nikolaev, 2024; Bai et al., 2023; Bybee, 2023; Jha et al., 2024). Examples include surveys and conceptual discussions of LLMs for economic research (Korinek, 2023) and applications that construct exposure measures, scores, or match-quality metrics from unstructured text (Eisfeldt, Schubert, and Zhang, 2023; Jha et al., 2024; Chen et al., 2024; Li et al., 2025).

Our setting raises a distinct measurement challenge: extracting structured policy attributes from long, heterogeneous government documents, where hallucination risk is salient. We therefore develop a multistage, verification-oriented pipeline. We decompose complex tasks into simpler sub-queries, separate text extraction from downstream classification, and require supporting excerpts to facilitate ex post checks against the source documents. Building on LLM-agent and iterative refinement ideas (Li et al., 2024), we combine outputs across models and incorporate expert review of error summaries used in subsequent refinement rounds. We validate the resulting classifications against human coders and benchmark performance against standard text-classification baselines (e.g., BERT and TF–IDF-based logistic regression and random forest), where the LLM-based approach performs better in our labeled evaluation sample.

The remainder of the paper is organized as follows. In Section 2 we review existing measures of industrial policies, discuss the advantages of LLMs in analyzing industrial policies, and provide details on applying LLMs in analyzing policy documents; we also describe several additional datasets of Chinese firms and politicians that we will use in combination with the industrial policy dataset; in Section 3, we establish the credibility of

[1] See Branstetter and Li (2023) for a comprehensive review on studies of China's industrial policies.

the LLM classifications; in Section 4 we present a detailed description of the landscape of China's industrial policies over the last twenty plus years using the data we extracted using LLMs; in Section 5 we document a set of facts about China's industrial policy over the past two decades, focusing on the economic and political foundations of sector choice and its evolution over time; finally in Section 6 we conclude.

2 Data and Measurement

2.1 Existing Measurement of Industrial Policy

The existing literature offers three main approaches to measuring industrial policy. The first approach relies on structured policy documents or specific policy shocks. Many studies rely on Five-Year Plans or other policy announcements to identify targeted industries. In China, national and provincial Five-Year Plans list "pillar" or "encouraged" industries prioritized for development. For example, Barwick, Kalouptsidi, and Zahur (2025) identify targeted sectors in provincial plans using keyword searches (e.g., "pillar industry"), while Cen, Fos, and Jiang (2024) code "encouraged industries" from national plans at the four-digit level and relate them to later subsidies; Chen, Li, and Xin (2017) adopt a similar strategy. Other studies exploit specific programs as shocks, such as the 2008 InnoCom initiative (Wei et al., 2023) or "Made in China 2025" (Li and Branstetter, 2024). Related designs appear in studies of industrial policies in other countries, including Juhász (2018) on the Napoleonic Blockade and Lane (2025) on South Korea's heavy and chemical industry (HCI) drive.

The second approach infers policy support from firm-level data. Aghion et al. (2015) construct city–industry measures from the Annual Survey of Industrial Firms (ASIF) data by aggregating subsidies, tax holidays, interest payments, and tariffs. Branstetter and Li (2023) study the subsidies received by listed-firms using CSMAR data; and DiPippo et al. (2022) compare industrial policy expenditures across countries using firm-level intervention indicators. Relative to plan-based measures or relying on specific policy shocks, firm-level proxies capture realized transfers to firms but can miss policies implemented primarily through non-fiscal instruments, such as policies on preferential land supply and labor inputs, consumer subsidies, procurement, or equity support.

The third approach codes industrial policy directly from government documents. Sinclair and Zhang (2025) scrape the State Council documents and identify industry-related policies using keyword-based rules. Recent text-based approaches extend this idea to larger and cross-country settings. Using the Global Trade Alert (GTA) database, Juhász et al. (2025) manually classify industrial policy entries and fine-tune a BERT model to code additional policy items. Goldberg et al. (2024) apply related methods in the semiconductor context, and Evenett et al. (2024) develop the New Industrial Policy Observatory (NIPO) to track industrial policy trends.

Our LLM-based approach differs from the existing measures in at least two dimensions. First, the existing measures often emphasize national or provincial policies and therefore understate city-level variations; however, in China, many interventions are designed and implemented at the city level, and firms mostly interact with city-level policies. Thus, studying local implementation, central–local pass-through, and policy diffusion requires a finer geographic resolution. Recent advances in LLMs make it feasible to extract such multidimensional attributes at scale from large, unstructured policy corpora. Second, most measures primarily recover *which* industries are targeted, but less often capture *how* policies are implemented—policy implementation tools, eligibility conditions, and stated objectives—because these details are typically dispersed throughout long documents (with Evenett et al., 2024 a notable exception based on intensive manual coding).

2.2 Classifying and Coding Policy Documents Using LLMs

We begin with the near universe of official government documents issued by China's central and local governments between 2000 and 2022. The core data come from PKULaw.com, an authoritative online platform hosted by Peking University Law School, which includes a broad range of policy-related documents—regulatory guidelines, policy announcements, and official statements from multiple levels of government. We supplement this source by scraping official government websites across administrative levels (central, provincial, and city-level), focusing on key ministries, departments, and affiliated agencies. The resulting dataset contains 3.28 million unique government documents spanning 2000–2022.

We use large language models (LLMs) to identify industrial policy documents under a consistent definition (outlined below) and to extract a set of policy attributes. For each document, we code the policy tone (supportive or regulatory/suppressive), the issuing authority, and the issuance date; we also identify targeted industries and map them to China's standard industry codes. In addition, we classify policy instruments into 20 categories grouped into five families: fiscal and financial tools; entry and regulation tools; input policy tools; demand-side tools; and supply-chain tools.[2] Guided by structured prompts, the LLMs transform unstructured policy text into a standardized dataset suitable for empirical analysis (see Appendix B for implementation details). This process converts lengthy raw texts into organized observations, enabling the analysis presented in Section 5.

Defining Industrial Policy. As a first step, we provide LLMs with a clear definition of industrial policy. Following Naughton (2021) and Juhász et al. (2025), we adopt a relatively narrow definition: industrial policy refers to *selective and targeted* government interventions intended to alter the sectoral structure of production toward sectors expected to deliver higher growth than would occur in a non-interventionist equilibrium. Specifically,

[2]We also extract information on policy objectives, organizational structures, incentive schemes, and inter-governmental citation networks, which are not analyzed in this paper.

a document qualifies as industrial policy if it meets four criteria:

1. The policy issuer must be a *government entity* at any level; texts involving only firms or non-governmental organizations are excluded.
2. The policy must contain *concrete government actions* rather than general statements or progress reports.
3. The policy must be explicitly *biased toward* (for or against) specific industries; broad macroeconomic programs without targeted sectors are excluded.
4. The policy must aim at *long-term* structural transformation rather than short-term stabilization, such as responses to COVID-19 or cyclical downturns.

Remark 1 An implication of Criterion 3 is that some broad, economy-wide reforms that may have heterogeneous effects on different sectors –for example, *Hukou* reforms benefit labor-intensive industries, WTO accession aids export-oriented sectors, and a carbon tax promotes green industries – would not be counted as industrial policies in this study. However, removing criterion 3 would essentially equate industrial policies with broad economic policies.

Policy Tone. Industrial policies can be supportive, regulatory, or suppressive; we refer to this dimension as the *policy tone*. Even under our narrow definition, an industrial policy need not be supportive. Supportive policies aim to promote sectoral development through measures such as innovation and upgrade support, recruitment and training of the workforce, improvements to the business environment, infrastructure investment, or lower entry barriers. Regulatory policies impose or strengthen standards and oversight (e.g., product and industry standards, environmental enforcement, market regulation, corporate supervision, and production safety), which may have longer-run implications. Suppressive policies seek to limit or phase out activities, for example, by restricting overproduction, eliminating outdated capacity, or tightening constraints on energy-intensive sectors.

Target Industry/Sector. Because our definition requires sector targeting, we identify the industries targeted in each policy at the four-digit level. We distinguish directly targeted industries from industries that are merely mentioned and from industries that may benefit only through spillovers. For example, a policy promoting electric vehicles may indirectly benefit battery producers; unless the document explicitly targets batteries, we do not classify the battery industry as directly targeted.[3] It is possible that multiple industries are targeted in one industrial policy document.

Policy Implementation Tools. A central task is to identify and categorize policy implementation tools. There is no widely accepted taxonomy of policy tools. The UNCTAD

[3]These spillover effects will be captured in our analysis when we incorporate input-output linkages in Section 5.1.

Multi-Agency Support Team (MAST) provides a coding scheme for trade-related measures, but it is designed primarily for *national-level trade* policy. We extend existing frameworks to encompass *domestic* industrial policies, especially at the local level, that are not necessarily related to international trade. We retain tools relevant to trade and international competitiveness—entry subsidies, import/export controls, financing tools, tax incentives, and fiscal subsidies—while adding instruments frequently used by sub-national governments: labor policies, preferential land supply, infrastructure investment, industrial funds, and policies that promote industrial clusters. In total, we have 20 policy tools, which we group into five families:

1. **Fiscal and Financial Tools:** credit and financing programs, tax incentives, equity support, and fiscal subsidies.
2. **Entry and Regulation Tools:** industrial funds, entrepreneurship promotion, investment policy, business-environment improvements, market-access rules, and trade-protection measures (e.g., import/export controls and non-tariff barriers).[4]
3. **Input Policy Tools:** labor policies (e.g., training and preferential *Hukou* policies), preferential land supply, infrastructure investment (e.g., transport and energy), policies promoting R&D and technology adoption, and environmental policies.
4. **Demand-Side Tools:** consumer subsidies, government procurement, and industrial promotion (e.g., exhibitions and trade fairs).
5. **Supply-Chain Tools:** policies supporting industrial clusters, and localization policies that encourage the use of local labor and suppliers.

This grouping provides a common language for summarizing the choice of policy tools across administrative levels and over time, and it facilitates comparisons of policy mixes across places, sectors, and stages of sectoral development.

2.3 Other Datasets

We merge the industrial policy database with rich firm-level data and a comprehensive dataset on provincial and city leaders to study political economic foundations and real effects.[5]

Firm Registration Data. We use the firm registration database from the State Administration for Market Regulation (SAMR), which covers the universe of registered firms in China. The database records a firm's location, industry (3-digit), registration date and exit date (if any), registered capital, ownership and equity change histories, and major shareholders. We aggregate the micro records to construct city–industry–year measures of

[4]Regulatory mechanisms can appear in both supportive and regulatory policies; we classify a policy as supportive when it relaxes existing rules or simplifies procedures.

[5]Summary statistics are reported in the Appendix Tables A4-A6.

entry (e.g., log new registered capital and counts of new registrations), exit, and the stock of registered capital.

Administrative Tax Records. To enrich our measure of firm performance, we also use the administrative enterprise income tax records from the Chinese State Administration of Taxation (SAT) for the years 2008-2020, covering a representative sample of more than 1 million firms from stratified sampling.[6] For tax collection and audit purposes, SAT collects firm-level records of tax payments, as well as other financial statement information used in tax-related calculations. The advantages of tax data are two-fold. First, it is representative with wide coverage. As administrative data, it is not subject to the potential measurement error problem of self-reported data. Second, it contains detailed historical information on a wide array of tax-related information about the firms, including their total production, sales, inputs, employment, etc. This allows us to examine the firms' *performance* in multiple dimensions.

Value-Added Tax Data. The third firm-level dataset that we use in our analysis is a unique dataset from the SAT that catalogs all Value-Added Tax (VAT) invoices issued by firms in mainland China from 2014 through 2018. It encompasses over 16.1 billion transactions among 18 million entities, providing a granular view of commercial exchanges within the Chinese economy. It includes detailed information on each transaction, such as the location of the registration and the 3-digit industry classification of the buyer and seller firms, the tax levied, and the transaction values. VAT filing is required for all transactions to claim input tax credits, and failures to comply are strictly punishable by law.[7] In Section 5, we use this dataset to construct upstream/downstream linkages and city-to-city trade flows to measure the extent of local protectionism and analyze its relationship with industrial policies.

Politicians Data. We manually collect all provincial and prefectural leaders, including party secretaries, governors, and mayors, who served between January 2003 and December 2019. For each city party secretary and mayor, we record their age, sex, birthplace, education, work experience, and factional ties. These data enable analysis of how leadership characteristics and networks shape industrial policy across regions and administrative levels.

[6] China's SAT is the equivalent of the Internal Revenue Service (IRS) in the United States.

[7] It is important to note that there are certain exceptions to transactions that fall outside the purview of the VAT invoice policy, for example, VAT invoices cannot be issued for transactions exempt from VAT, sales to consumers and several other specified conditions; also it supports small businesses by providing VAT exemptions based on sales thresholds while ensuring that larger enterprises engage in compliant invoicing practices.

3 Validation and Benchmarking

We assess the reliability of LLM-based classifications and extracted attributes using two complementary exercises. First, we conduct an internal validation that compares document-level LLM labels with independently produced human labels for a random subsample of the policy documents; we also benchmark the LLM performance against standard text-classification baselines (random forest, BERT and logistic regressions). Second, we conduct an external validation that asks whether the resulting measures of industrial policies predict subsequent firm entry and outcomes.

3.1 Internal Validation against Human Labels

We recruited 20 research assistants (RAs) to label 2,000 randomly sampled documents from the corpus described in Section 2.2. Each RA labeled 300 documents, and each document was independently labeled by three RAs. We provided three training sessions and a detailed coding manual to the RAs, and they were instructed to apply the definition of industrial policy in Section 2.2: the issuer must be a government entity; the document must specify concrete policy actions; the intervention must be sector-targeted; and the stated goal must be structural rather than short-term stabilization.

It is useful to note that, perhaps not surprisingly, human coders do not always agree with their labels: for 19.25% of the policy documents, the three RAs did not all agree on whether a policy document is an industrial policy document. We use two options to aggregate the three RAs' labels: first, a document is classified as an industrial policy if at least two of the three RAs label it as such; second, a more stringent option is that a document is classified as an industrial policy if all three RAs label it as such. Using the RA label as the "ground truth," Table 1 reports the confusion matrix for the binary classification of the LLM. Panel A reports the statistics for the full sample where the RA majority is used as the ground truth: the LLM achieves an accuracy rate of 92.6%, a precision rate of 80.6%, a recall rate of 78.4%, and an F1 score of 79.5%. Panel B reports the statistics for the subsample in which all three RAs agree: the LLM achieves an accuracy rate of 97.1%, a precision rate of 87.2%, a recall rate of 90.7% and an F1 score of 89%. Thus, LLM labels seem to perform better against human labels when they all agree.

[Table 1 about here]

3.2 External Validation: Predictive Power for Firm Behavior

Internal benchmarking establishes agreement with human coders; for economic applications, however, we also require that the extracted measures anticipate real outcomes. Therefore, we test whether exposure to industrial policies constructed from LLM classifications forecasts subsequent firm entry dynamics and outcomes, using the datasets in

Section 2.3. It is important to emphasize that the purpose here is predictive (e.g., content validity), not causal.

We focus on two families of outcomes. First, using administrative tax records, we test whether firms in targeted city–industry cells exhibit changes consistent with policy support, e.g., higher subsidies, greater tax deductions, or improved access to long-term financing. To account for differential effects by policy tones, we separately construct indicators for supportive policies, $Policy^{+}_{cst}$, and for regulatory/suppressive policies, $Policy^{-}_{cst}$, where $Policy^{+}_{cst} = 1$ (respectively, $Policy^{-}_{cst} = 1$) if and only if the city c issued a supportive (respectively, regulatory/suppressive) industrial policy targeting industry s in year t. We estimate event-study-style specifications with two leads and two lags of policy indicators to control for inter-temporal correlation of policy (where negative values of l correspond to leads and positive values to lags):

$$Y_{fcst} = \beta_0 + \sum_{l=-2}^{2} \beta_l^{+} \times Policy^{+}_{cs(t-l)} + \sum_{l=-2}^{2} \beta_l^{-} \times Policy^{-}_{cs(t-l)} + \delta_{cs} + \gamma_t + \epsilon_{fcst} \quad (1)$$

where Y_{fcst} captures firm-level financial outcomes, including: (i) tax deduction rate (total tax exemptions divided by pre-tax revenue); (ii) log of the subsidies received; and (iii) an indicator of long-term debt. We include city by industry fixed effects δ_{cs} and year fixed effects γ_t.

Figure A2 reports the estimated coefficients of β_l^{+} and β_l^{-} with the corresponding 95% confidence intervals. It shows that supportive policies predict higher subsidies, higher tax deductions, and greater access to long-term debt, whereas regulatory/suppressive policies exhibit weaker or opposite correlations.

Second, we examine whether policy exposure measures predict firm entry responses. Using the firm registration data, we aggregate outcomes to city–industry–year cells (3-digit industries).[8] Our main entry outcome is log new registered capital at the city–industry–year level. We estimate event-study-style specifications with four leads and six lags of supportive and regulatory/suppressive policy exposure:

$$Y_{cst} = \sum_{l=-4}^{6} \beta_l^{+} \times Policy^{+}_{cs(t-l)} + \sum_{l=-4}^{6} \beta_l^{-} \times Policy^{-}_{cs(t-l)} + \delta_{cs} + \gamma_t + \epsilon_{cst}. \quad (2)$$

We include city by industry fixed effects δ_{cs} and year fixed effects γ_t.

Panel A of Figure A3 reports the estimated coefficients of β_l^{+} and β_l^{-} with the corresponding 95% confidence intervals. It shows that supportive policies predict higher entry around the year of issuance, with effects that attenuate over subsequent years; in contrast, regulatory/suppressive policies predict a lower entry, with the decline materializing with

[8] We extract targets at the 4-digit level and aggregate to 3-digit industries for the baseline empirical analysis to align with firm and VAT datasets where industries are available at the 3-digit level.

lags.

Comparing with Five-Year Plans. A natural question is whether the predictive content of our LLM-based measures merely replicates the information already captured by traditional indicators derived from the "pillar" or "encouraged" industries in China's Five-Year Plans. To address this, we re-estimate the equation while absorbing all variations associated with the Five-Year Plans by including city-by-industry-by-plan-cycle fixed effects. This specification nets out the Five-Year Plans designations and other persistent development priorities within each five-year plan cycle.

Panel B of Figure A3 plots the estimated coefficients of β_l^+ and β_l^- with the corresponding 95% confidence intervals. Even after controlling for these highly saturated fixed effects, the LLM-based measure continues to exhibit strong and statistically significant predictive power on firm entries. The results imply that, within a given five-year plan cycle, our measure captures meaningful higher-frequency and sub-national variations that traditional Five-Year Plans based indicators of industrial policies miss.

3.3 Benchmark against Random Forest, BERT and Logistic Regressions

We also benchmark the performance of LLM-classifications with BERT models, the random forest model, and logistic regressions with different regularizations on the same human-labeled sample. We partition the 2,000 documents into 1,400 for training, 300 for validation, and 300 for testing, and average the results over three random iterations. The "layer-optimized" BERT fine-tunes the classification layers only, whereas the "fully-optimized" BERT fine-tunes the full network. We conduct extensive hyperparameter searches to optimize each benchmark model, as detailed in Appendix D.

Table A1 shows that LLM performance dominates that of the other models on all metrics. Relative to all other models, it achieves higher recall without sacrificing precision, which translates into a materially higher F1 score.

Remark 2 When coding complex and long text documents, there will be measurement errors regardless of whether we use human RAs or hallucination-robust LLMs. However, processing more than 3 million complex documents is feasible only with LLMs due to their scalability. The validation results presented in this section make us confident in using the LLM outputs to study China's industrial policies.

4 Landscape of China's Industrial Policies

Having validated the performance of LLM classifications, we now present an overview of China's industrial policy landscape over the last 20+ years using the LLM-constructed

industrial policy database. We first summarize policy volumes over time and across administrative levels, then characterize the distribution of policy tones, targeted industries, and policy implementation tools; finally, we evaluate the effectiveness of different policy tools in stimulating new firm entries and in stimulating firm productivity, and show interesting evolution of policy tools over the life-cycle of the targeted industries.

4.1 Overview

Table 2 reports that LLMs classified 595,957 of them as industrial policies, representing 18.16% of the 3.28+ million unique government policy documents. It also reports summary statistics by government level. By government level, 19.58% of central-government documents are classified as industrial policies; the corresponding shares are 18.83% for provincial governments and 17.27% for city governments. In terms of counts, central, provincial and city governments represent 14.16%, 39.46% and 42.58% of all industrial policy documents, respectively.[9] This distribution underscores the importance of local policies in the overall landscape of China's industrial policies.

[Table 2 about here]

Panel A of Figure 1 plots the annual volume of all government documents (solid line, left axis), the subset classified as industrial policies (dashed line, left axis), and the share of industrial policies among all documents (dotted line, right axis). Both the total number of documents and the number of industrial policy documents increased steadily from 2000 to 2022, while the share of industrial policy remains relatively stable over time.

[Figure 1 about here]

Spatially, industrial policy activity is more concentrated in wealthier and more industrialized regions in South and East China, notably Guangdong, Jiangsu, and Zhejiang, and is less dense in the Central and Western regions.[10]

4.2 Policy Tones

Panel B of Figure 1 plots, by issuing government level, the share of industrial policy documents with a supportive tone over time. In the full sample, 69.6% of industrial policy documents are supportive, 28.9% are regulatory and 1.5% are suppressive. The supportive share varies systematically across government levels: 56.5% for central-government policies,

[9] County or district level policies represent 3.79% and township level 0.01%. Due to the significantly smaller volume of these lower-level government policies , our analysis below focuses only on the central, provincial, and city-level industrial policies.

[10] Figure A4 maps the spatial distribution.

69.3% for provincial policies, and 74.4% for city policies. Over time, supportive policies become more prevalent at all levels. The central government issues relatively more regulatory policies, though the center–local gap in tone composition narrows after 2013, suggesting a more proactive role of the central government in the practice of industrial policy. These facts also imply that restricting attention to supportive policies, as almost all existing literature does, omits a sizable regulatory/suppressive component of the industrial policy landscape. Indeed, to the extent that industrial policies are aimed at affecting the long-term structure of the economy, both supportive and regulatory/suppressive tools should be part of the policy mix.

4.3 Distribution of Target Industries

Figure 2 summarizes how the sectoral composition of the targeted industries evolves over time. Panel A aggregates the targeted industries to the 2-digit level. Manufacturing accounts for the largest share of targeted industries (more than 30%), while over time the share of policy for agriculture declines and the share for services increases. Panel B focuses on manufacturing and plots the share of industrial policies targeting high-skill manufacturing industries within the manufacturing sector. It shows that the share targeting high-skill manufacturing increases steadily, even as the overall share of industrial policies targeting the manufacturing sector edges down slightly (as seen in Panel A), consistent with a shift in the target toward technologically more advanced segments.

[Figure 2 about here]

4.4 Policy Implementation Tools

As discussed in Section 2.2, we classify the policy implementation tools into five broad families: fiscal and financial tools; entry and regulation tools; input policy tools; demand-side tools; and supply-chain tools. Multiple tools can be deployed in a single document. Table 3 reports the share of tool usage, namely, the fraction of industrial policy documents issued by each issuing government level that contain a given policy tool.

Several policy tools are widely used at all levels. Fiscal subsidies are the most common instrument, appearing in about 49% of policies. Although this aligns with the view that industrial policy often relies on subsidies, it also implies that more than half of industrial policy documents do not involve fiscal subsidies, thus cautioning against proxying industrial policy using fiscal subsidies alone. Other frequently used tools include market access and regulation (43%), labor policy (28%), R&D and technology adoption (27%), and infrastructure investment (24%).

[Table 3 about here]

The choice of policy tools also varies across administrative levels. The central government relies most heavily on market access and regulation (53%), followed by fiscal subsidies (26%), R&D and technology adoption (20%), labor policies (17%), and trade protection (17%). City governments place greater weight on fiscal subsidies (56%) and also use market access/regulation (52%), labor policies (33%), R&D and technology adoption (29%) and infrastructure investment (29%) more intensively; provincial governments lie in between and are closer to cities than to the center. The pattern is consistent with a division of roles in which the central government sets access rules and standards, while sub-national governments deploy a broader mix of financial and non-financial instruments in the context of inter-jurisdictional competition. Local intensity is also evident in the higher usage of policy tools such as labor policies (33% at the city level versus 17% at the central level), infrastructure investment (29% versus 13%), and industrial cluster promotion (24% versus 9%).

Remark 3 The Chinese government industrial policy documents do not articulate the market failure(s), e.g., agglomeration effects (economies of scale), coordination failure, externalities, etc., that the policies are motivated to overcome. However, it is worth emphasizing that some frictions that the policies aim to address result from other existing government policies, instead of being intrinsic to the market. For example, labor policies tend to overcome the labor market frictions induced by China's *Hukou* policy, a residence policy that restricts migrant workers from being able to access the public goods in the cities they work. Some of the finance and subsidy policy tools help address the frictions of financial markets that are specific to China's state-driven financial system, e.g., Chinese banks' preference toward state-owned enterprises.

Policy Tools and Firm Outcomes. Next, we relate policy tools to firm outcomes, including entries and TFP. Focusing on supportive policies, we estimate a city–industry–year regression of entry on contemporaneous policy tool exposures:

$$Y_{cst} = \sum_{k} \beta_k \times Tool_{kcst} + \delta_{cs} + \gamma_t + \epsilon_{cst}, \tag{3}$$

where Y_{cst} is the log number of new firm registrations in industry s, city c, and year t; $Tool_{kcst}$ is an indicator of whether the policy tool k is deployed in supportive policies targeting industry s in city c, year t; and δ_{cs} and γ_t are respectively the city $\times$ industry and the year fixed effects.

Panel A of Figure 3 plots the degree of estimated association of various policy tools, i.e., the β_ks, with new firm entries. It shows that new firm entry is most strongly associated with fiscal subsidies, infrastructure investment, preferential land supply, and industrial cluster promotion policies; access to credit and finance, labor policies, market access/regulation, and industrial promotion are also positively associated with new firm entry, but to a lesser

degree. In contrast, environmental policies, trade-protection measures, and equity support are negatively associated with entry, consistent with these instruments being deployed in settings with higher compliance burdens, tighter barriers to entry, or support tilted toward incumbent or larger firms, although the specification is only descriptive.

[Figure 3 about here]

We then examine productivity using a regression specification similar to Eq.(3), except that the outcome variable is the firm-level TFP. Several mechanisms operate in opposite directions. For example, on the positive side, entry subsidies and low-cost land can intensify competition by shifting out low-productivity firms; demand-side tools can raise revenues and, through market selection, shift shares toward more productive firms; and cluster promotion policies can help coordinate the emergence of supply chains and improve productivity for all firms; in contrast, easier entry can draw in low-productivity firms, and the resulting overcapacity can crowd out resources and lower the productivity of all firms.

Panel B of Figure 3 plots the estimated degree of association of various policy tools, i.e., the β_ks, with a firm's TFP. It shows that the R&D and technology adoption policy tool is most strongly positively associated with the TFP of firms, followed by policy tools to promote the industrial cluster, highlighting the role of reduced input–output frictions; and fiscal subsidies, plausibly directly via revenue effects. Furthermore, demand stimulation, including consumer subsidies and government procurement, preferential land supply, and capital-deepening tools, including industrial fund and investment policies, also tend to increase TFP, albeit to a lesser degree. However, promoting entrepreneurship is negatively related to TFP, plausibly reflecting the influx of smaller, less productive entrants and the resulting resource crowding out; labor policy is also negatively related to TFP, echoing existing evidence that relaxing labor mobility constraints can lower productivity by encouraging firms to switch to labor intensive technology with lower TFP (Imbert et al., 2022). The other tools are not significantly related to TFP in these specifications.

Dynamics of Policy Tool Choices over the Life-cycle of Targeted Industries. Finally, we ask whether local governments adjust the choice of policy tools over the local life cycle of a targeted industry. For each city–industry pair, we construct *duration* as the number of years since the first city-level industrial policy targeting the industry was issued. We then relate tool usage in a given policy document to duration, controlling for year and city × industry fixed effects. We estimate the following Poisson pseudo-maximum likelihood (PPML) estimator (Silva and Tenreyro, 2006; Silva and Winkelmann, 2024), which accounts for the large share of zeros in the dependent variable:

$$\mathbb{1}(Tool_{ikcst}) = \exp\Big[\beta^k \, Duration_{cst} + \gamma_t + \delta_{cs}\Big] + \epsilon_{ikcst}, \tag{4}$$

where $\mathbb{1}(Tool_{ikcst})$ indicates whether the industrial policy document i uses tool k when targeting industry s in city c in year t, and $Duration_{cst}$ is years since industry s was first

targeted in city c. With city $\times$ industry fixed effects δ_{cs}, identification comes from within-(c, s) changes in tool choice depending on the industry's local life cycle, net of common year shocks γ_t. We estimate Eq.(4) separately for each tool k.

Figure 4 plots the estimated $\hat{\beta}^k$ for all 20 policy tools: a policy tool with a negative (respectively, positive) coefficient indicates that it tends to be used more frequently in the early (respectively, later) stages of the industry being locally targeted. The results suggest systematic life-cycle patterns. Toward the beginning of an industry to be targeted, policies rely more on government-led and entry-oriented demand instruments, including investment policy, tax incentives, fiscal subsidies, preferential land supply, market access/regulation, and government procurement. As duration increases, the choice of tools changes toward upgrading and selection, with greater use of technology R&D and technology adoption, industrial funds and equity support, labor and talent development, environmental policies, and cluster promotion; on the demand side, the policy tool changes from procurement to industrial promotion and consumer-oriented stimulation. Taken together, the evidence is consistent with local governments adjusting the mix of policy tools as the targeted industries mature.

[Figure 4 about here]

5 Political Economy of China's Industrial Policies

This section studies the political economy of China's industrial policy over the past two decades. We first examine how local governments select targeted sectors and how these choices relate to local economic fundamentals and to upper-level priorities. We then trace how the relative importance of these determinants evolves over time and document a marked increase in top-down political alignment after 2013. Next, we examine the mechanisms behind this shift and explore its consequences. Throughout this section, we focus on supportive policies.

5.1 Political-Economic Determinants of Targeted Industries

China's Regional Decentralized Authoritarianism (RDA) combines centralized political authority with substantial decentralization in the implementation of economic policies, with tournament-style promotion incentives for local officials (Xu, 2011). In this environment, the choice of which industries to target in local governments' industrial policy can reflect an intricate balance of local economic fundamentals and top-down policy priorities. Therefore, we examine the extent to which city's decisions about which sectors to target are driven by economic considerations – measured by the city's revealed comparative advantage (and absolute advantage) in the industry, the industry's position in the production-network, and the coordination along upstream and downstream links, and by upper-level governments'

policy portfolios that capture top-down political initiatives. We also investigate whether the balance between these forces changes over time.

Economic Primitives. A central question in the industrial policy literature is whether governments target sectors that align with existing comparative advantage, i.e. "comparative advantage following," or instead support sectors that currently lack comparative advantage but may develop it over time, i.e. "comparative advantage defying" (Calcagno et al., 2015; Rodriguez-Clare, 2007). The former view argues that interventions should build on the current strengths of the economy, since firms operating within comparative advantage are more likely to be profitable and thus sustain growth (Lin and Chang, 2009; Lin, 2015). The latter view emphasizes infant-industry considerations: supporting nascent activities can induce learning and productivity growth that eventually generates comparative advantage (e.g. Redding, 1999; Melitz, 2005; Greenwald and Stiglitz, 2006). Recent cross-country evidence suggests that, in the recent wave of industrial policies, governments tend to support industries in which they already have revealed comparative advantage (Juhász et al., 2025; Evenett et al., 2024).

Following Balassa (1965) in spirit, using cumulative registered capital from the firm-registration database, we create capital-based revealed comparative advantage (RCA) and absolute advantage (AA) measures as follows:[11]

$$RCA^{p}_{cst} = \frac{Capital_{cst}}{\sum_s Capital_{cst}} \Bigg/ \frac{\sum_{c' \in p(c)} Capital_{c'st}}{\sum_s \sum_{c' \in p(c)} Capital_{c'st}}, \tag{5}$$

$$AA_{cst} = \frac{Capital_{cst}}{\sum_s Capital_{cst}}. \tag{6}$$

where RCA^{p}_{cst} captures whether city c is more specialized in industry s than other cities in the province, and AA_{cst} reflects the absolute importance of industry s in city c.[12]

Local Supply-chain Structure and Coordination. Using transaction-value information, we compute trade-value weighted degree centrality in the local supply-chain network within the city and the province, respectively:

$$Centrality^{c}_{cs} = \sum_j Trade^{c}_{sj}, \tag{7}$$

$$Centrality^{p}_{p(c)s} = \sum_j Trade^{p}_{sj}, \tag{8}$$

[11] Balassa (1965)'s original RCA index is concerned with assessing a country's comparative advantage in global exports, whereas our modified measure intends to evaluate a city's relative strength in domestic production.

[12] Including a national-level RCA analog produces similar, but weaker associations once provincial RCA is included.

where $Trade^{c}_{sj}$ and $Trade^{p}_{sj}$ are the total value of trade between firms in industry s and j within the city and provincial supply-chain networks, respectively, calculated from our VAT data for years 2017-2018.

To capture policy coordination along supply chains, we weight contemporaneous supportive industrial policies in linked upstream and downstream industries using national-level input and output shares, specifically, we use the 2017-2018 VAT data to calculate $Input_{sj}$, the share of the inputs of industry s purchased from industry j; and $Output_{sj}$, the share of the output of industry s sold to industry j. In addition, we define $Policy^{+}_{cst}$=1 if there is at least one supportive industrial policy document issued by city c targeting industry s in year t, where we collapse the targeted industries to the 3-digit level.[13] We define the supportive industrial policy exposures of the upstream and downstream of sector s as:

$$Policy^{Up,+}_{cst} = \sum_{j \neq s} Policy^{+}_{cjt} \cdot Input_{sj}, \tag{9}$$

$$Policy^{Down,+}_{cst} = \sum_{j \neq s} Policy^{+}_{cjt} \cdot Output_{sj}, \tag{10}$$

The mean of $Policy^{+}_{cjt}$ across all city–industry–year cells is about 8%, that is, on average 8% of city–industry–year cells are targeted with supportive industrial policies. Again, to account for the large share of zeros in the dependent variable, we estimate the following PPML:

$$\begin{aligned} Policy^{+}_{cst} = \exp\Big[\lambda_1 RCA^{p}_{cs(t-1)} + \lambda_2 AA_{cs(t-1)} + \lambda_3 \log(Centrality^{c}_{cs}) + \lambda_4 \log(Centrality^{p}_{p(c)s}) \\ + \lambda_5 Policy^{Down,+}_{cst} + \lambda_6 Policy^{Up,+}_{cst} + \delta_c + \eta_s + \gamma_t\Big] + \epsilon_{cst}, \end{aligned} \tag{11}$$

where δ_c, η_s and γ_t stand for fixed effects of city, industry and year, respectively. We also report results for specification with more saturated fixed effects, city-by-year and industry-by-year fixed effects in Column (3), and city-by-industry fixed effects in Column (4). Notice that the local fundamental variables $RCA^{p}_{cs(t-1)}$ and $AA_{cs(t-1)}$ enter with one year lag to avoid mechanical contemporaneous links.

[Table 4 about here]

Table 4 reports the regression results. Across specifications, we robustly find that cities are more likely to target industries in which they have a higher revealed comparative advantage, $RCA^{p}_{cs(t-1)}$, and a larger local presence, $AA_{cs(t-1)}$, though the impact of an industry's RCA in the province dwarfs that of the industry's AA. The estimates in Column (3) implies that a one-standard-deviation increase in $RCA^{p}_{cs(t-1)}$ is associated with a 16.7% higher probability that the industry is targeted in the city (0.0172×9.7), and a one-standard-deviation increase in $AA_{cs(t-1)}$ is associated with a 1.5% higher targeting probability (1.137×0.013).

[13] Our results are robust to alternative policy measure using the count of individual policy documents targeting specific industries.

Table 4 also shows that the industry's position in the supply-chain network at the city and provincial levels, as proxied by the centrality measures $Centrality^{c}_{cs}$ and $Centrality^{p}_{p(c)s}$ defined in Eq.(7), are positively associated with the industry being targeted by the city government. In addition, an industry is more likely to be targeted by local industrial policies when its upstream and downstream industries are also supported, consistent with coordinated targeting along production networks.

Remark 4 A potential concern is serial correlation: earlier targeting may itself raise subsequent RCA^{p} and AA, in which case the association between targeting and advantage could partly reflect policy-induced changes rather than deliberate selection. To address this, we re-estimate Eq.(11) using only the *first* year an industry is ever targeted in a city and lagged advantage measures from the prior year. The associations remain strong (Appendix Table A7).

Top-Down Political Alignment. We next turn to political-economy determinants of the local governments' choice of which industries to target, focusing on *top-down* pass-through: the extent to which city-level sectoral targets mirror those of the provincial and the central governments. We estimate specifications that relate a city's targeting indicator to upper-level targeting and allow the pass-through to vary with city characteristics:

$$Policy^{+}_{cst} = \exp\Big[\beta_1 Policy^{p+}_{p(c)st} + \beta_2 Policy^{p+}_{p(c)st} \times \mathbf{X}_{ct} + \beta_3 Policy^{n+}_{st} + \beta_4 Policy^{n+}_{st} \times \mathbf{X}_{ct} + \beta_5 \mathbf{X}_{ct} + \delta_{cs} + \gamma_t\Big] + \epsilon_{cst}, \quad (12)$$

where $Policy^{p+}_{p(c)st}$ and $Policy^{n+}_{st}$ indicate whether city c's province, $p(c)$, and the central government, respectively, support industry s in year t; and $\mathbf{X}_{ct}$ is a vector of city characteristics including (i) (log) GDP to proxy development and administrative capacity; (ii) (log) number of cities in the province to proxy the intensity of promotion competition within the province;[14] and (iii) indicators for whether the city party secretary or mayor has personal ties to provincial officials (shared hometown, college, or previous workplaces). All specifications include city-by-industry fixed effects δ_{cs} and year fixed effects γ_t.

Table 5 reports the results. Column (1) provides the baseline top-down pass-through estimates and shows that cities are more likely to support industries that are targeted by the provincial and central governments, with substantially stronger alignment vis-à-vis the province. Columns (2)–(4) explore heterogeneity. Column (2) shows that pass-through is weaker in richer cities, suggesting that richer cities have more fiscal space to target industries that are more tailored to their local comparative advantages. Column (3) shows that the province-to-city pass-through is stronger in provinces with more peer cities, consistent with

[14]This proxy is used in Lü and Landry (2014). The idea is that, in a given province, a larger pool of city leaders competing for promotion intensifies relative performance incentives.

the notion that stronger promotion competition leads to more alignment with the higher-level government guidance. Column (4) shows weaker pass-through when local leaders have personal ties to upper-level officials, consistent with greater informal political capital, and hence more autonomy in tailoring the local industrial policy targets.

[Table 5 about here]

Together with the results reported in Table 4 on the impacts of local economic primitives, these estimates indicate that the local choices of which industries to support with their industrial policies reflect both economic considerations (comparative/absolute advantage, network position, and supply-chain coordination) and top-down political alignment shaped by local economic development, career incentives, and connections. We next examine how the balance between these forces evolves over time and how the post-2013 shift toward centralization changes the local choices of targeted industries and its consequences.

5.2 Recentralization After 2013: Evidence and Mechanisms

Section 5.1 shows that both local economic fundamentals and political forces predict local sectoral targeting. We now ask how the *relative* importance of these forces has evolved over time and whether the data exhibit a discrete shift toward greater top-down alignment, i.e., recentralization, after 2013.

5.2.1 Evidence of Re-Centralization After 2013

To make comparisons across places and years, we construct summary measures of the alignment of the local choices of targeted industries with (i) the upper-level government's priorities; and (ii) the revealed comparative advantage of the local industrial structure. Specifically, let $\mathbf{P}_{ct}$ denote the sectoral target vector of the industrial policies of city c in year t, where each element is a 0/1 indicator for whether a given industry is targeted by a supportive industrial policy, and let $\mathbf{P}_{p(c)t}$ denote the corresponding sectoral target vector of the industrial policies of the province $p(c)$ where city c belongs to. We measure the *political alignment* of the industrial policies of the city c with those of the provincial government by the cosine similarity of $\mathbf{P}_{ct}$ and $\mathbf{P}_{p(c)t}$:

$$Similarity_{ct}^{p} = \frac{\mathbf{P}_{ct} \cdot \mathbf{P}_{p(c)t}}{\|\mathbf{P}_{ct}\| \, \|\mathbf{P}_{p(c)t}\|} \tag{13}$$

Similarly, we construct the *economic alignment* of industry policies in a city with its local industrial structure by the cosine similarity of its sectoral target vector $\mathbf{P}_{ct}$ with its sectoral revealed comparative advantage vector, $\mathbf{R}_{ct}$, which is the vector of the sectoral capital-based RCA measure as described by Eq.(5):

$$Similarity_{ct}^{\text{RCA}} = \frac{\mathbf{P}_{ct} \cdot \mathbf{R}_{ct}}{\|\mathbf{P}_{ct}\| \, \|\mathbf{R}_{ct}\|} \tag{14}$$

Both similarity measures lie in $[0, 1]$.

Figure 5 plots the time series of the city averages of $Similarity^{p}_{ct}$ and $Similarity^{\text{RCA}}_{ct}$. From the early 2000s through 2012, neither series displays a pronounced trend. However, since 2013, the two measures have moved in opposite directions: the top-down political alignment $Similarity^{p}_{ct}$ increases while the alignment with the local revealed comparative advantage $Similarity^{\text{RCA}}_{ct}$ declines. In other words, from 2013 onward, the portfolios of sectors supported by Chinese cities' industrial policies on average move closer to the sectors supported by their provincial governments, but diverge from the cities' local revealed comparative advantages.[15]

[Figure 5 about here]

To assess whether the increasing political alignment $Similarity^{p}_{ct}$ systematically attenuates the role of more broad economic factors in predicting the cities' choices of targeted industries, we re-estimate Eq.(11) allowing the effects of the economic primitives to vary with $Similarity^{p}_{ct}$. Table A8 reports the results. The estimated coefficients for the interactions of $Similarity^{p}_{ct}$ with $RCA^{p}_{cs(t-1)}$, $AA_{cs(t-1)}$, as well as the industrial policy exposures of the upstream and downstream sectors $Policy^{Up,+}_{cst}$ and $Policy^{Down,+}_{cst}$, are negative and statistically significant; and the coefficient of the interaction with *city*-level network centrality $Centrality^{c}_{cs}$ is also estimated to be negative, indicating that economic predictors have weaker predictive content in cities that are more vertically aligned. In contrast, the estimated coefficient of the interaction with *province*-level centrality $Centrality^{p}_{p(c)s}$ is positive and significant, consistent with the amplification of provincial priorities in more top-down aligned cities.

5.2.2 Potential Mechanisms Driving the Re-Centralization

Our finding that there has been a re-centralization in China's economic policy-making since around 2013 is consistent with the recent literature (Fang and Zhang, 2025; Zhou et al., 2021; Bo, 2020; Lee, 2017; Luo, Wang, and Yang, 2025). We next examine the potential mechanisms underlying the re-centralization. We zoom in on two sets of forces. On the center's side, we show that the directives become more detailed, stronger in tone and expand to cover a wider set of sectors; on the local side, we show that the incentives to comply with the upper-government directives increase as cities become more fiscally constrained and as political promotions after 2013 place greater weight on alignment with upper-level priorities.

[15] It should be noted that there was a mild re-balancing of these measures after 2018: the political alignment remains elevated but stabilizes, while the alignment with local RCAs has modestly recovered.

Central Directives Became Stronger and Broader. We first characterize how the tones of the government policy directives have evolved over time. For each industrial policy document, we construct two text-based indices: *concreteness* and *strength*. Using the full text, the LLMs extract information on implementation arrangements (e.g., accountability provisions, organizational support, and resource commitments) and then score (i) how specific and measurable the stated objectives are (concreteness) and (ii) how binding the document describes the terms of enforcement and accountability (strength).[16] We standardize the resulting measures and report them on a 0–100 scale.

[Figure 6 about here]

Figure 6 plots the evolution of these two indices by issuing-government level. At the central level, both indices were relatively flat prior to 2013 and then increased sharply by roughly ten points on the scale of 0–100; in contrast, the provincial and city-level indices drifted modestly down. The narrowing center–local gap suggests that the central government industrial policy directives become more detailed and stronger relative to the sub-national documents, potentially reducing the scope for local discretion in their choice of targeted sectors.[17]

Second, we show that the *breadth* of sectors covered by the industrial policy documents issued by the central government has increased significantly since 2013. When the central government's industrial policy directives concentrate on a relatively small set of strategic industries, local governments have more discretion, both fiscally and politically, to allocate resources to support industries outside the prioritized set of the central government. However, when central initiatives expand to cover a broader range of sectors, the local governments' fiscal and administrative capacity would be strained, especially in cities with more limited own revenue sources. This would in turn incentivize the local government to support the industries prioritized by the central and provincial governments because the local governments could receive transfers from the upper-level governments for supporting industries the upper-level governments prioritize. To account for the differences in the quantity of supportive documents targeting industries, we measure the breadth of industrial policy at the national level based on the Herfindahl-Hirschman Index (HHI):

$$HHI_t^n = \sum_s \left[w_{st}^n\right]^2, \tag{15}$$

where $w_{st}^n = \#Policy_{st}^{n+} / \left(\sum_{s'} \#Policy_{s't}^{n+}\right)$ denotes the share of supportive policy documents targeting sector s among all supportive industrial policy documents issued by the central government in year t. Note that if all supported sectors have the same share, i.e. $w_{st}^n = 1/N$

[16] See Section C.4 for details.

[17] Interestingly, the upward trend of the central government's industrial policy directives somewhat reversed after 2018, consistent with the modest re-balancing toward a higher economic alignment vis-à-vis top-down political alignment, as documented in Figure 5.

where N is the number of supported sectors, then $HHI_t^n = 1/N$; thus, lower values of HHI_t^n indicate a wider sector portfolio targeted by the central level industrial policy.

In Figure 7, Panel (A) plots the evolution of HHI measures HHI_t^n from 2000 to 2020, showing that the breadth of central policy coverage decreased sharply after 2013, though it reversed somewhat after 2018. Panel (B) reports the corresponding (average) HHI measure at the city level, which also showed a significant decline after 2013, consistent with an increasing (average) number of supported sectors in the local industrial policy portfolio (Panel C). Thus, as the center articulates a wider set of priorities, local governments align by extending their own policy coverage. The post-2013 decline in city-level HHI coincides with the increase in top-down political alignment, consistent with the interpretation that the additional breadth at the local level is disproportionately filled by upper-level priorities rather than by distinctive local targeting. Indeed, Panel (D) presents the binscatter plot that relates the city-level policy HHI with the city's policy political alignment with the provincial government, as proxied by $Similarity_{ct}^p$ as defined in Eq.(13), after absorbing year fixed effect. It shows that cross-sectionally, cities with lower policy HHIs tend to exhibit higher alignment with provincial level policy portfolios.

[Figure 7 about here]

Local Fiscal Incentives. We next examine the potential mechanisms that link the post-2013 shift in the concreteness, strength, and breadth of the central government's industrial policy to the local governments' choice of targeted industries. The first channel is a *fiscal* channel. Implementing industrial policies to support a target sector costs fiscal resources. If the city government chooses a target sector that is among those supported by the upper-level governments, part of the costs could be paid for by the transfers from the upper-level governments; on the other hand, if the city targets a sector that is aligned with local RCA but not among those supported by the upper-level governments, then the local government must bear all of the costs. When central directives become broader and more concrete, local compliance becomes more observable; as a result, cities that rely more heavily on intergovernmental transfers in their budget may have stronger incentives to align with upper-level portfolios, whereas cities with greater own-source revenue, mainly land sales revenue, are more able to target sectors that confer local RCA but outside of the upper-governments' portfolio.[18] We construct two alternative measures of fiscal capacity: (i) log total land sales revenue, with a higher value indicating better fiscal condition; and (ii) a transfer ratio, defined as $(Fiscal\ Revenue - Total\ Taxes)/(Fiscal\ Revenue + Land\ Sales\ Revenue)$, which captures the city's reliance on intergovernmental transfers from upper-level governments (i.e., *Fiscal Revenue-Total Taxes*). The fiscal channel implies that *ceteris paribus*, worsening fiscal conditions of local government can be a force to drive more top-down alignment.

[18] In China, local government owns urban land and can sell land-use rights to raise revenues. Land sales revenues, on average, accounted for close to 40% of local governments' total revenues. See Fang et al. (2016) for details.

We test this channel by relating the city-level top-down alignment measure $Similarity^{p}_{ct}$ to the city's fiscal capacity, while allowing the relationship to change after 2013:

$$Similarity^{p}_{ct} = \beta_1 \mathbf{X}_{ct} + \beta_2 \mathbf{X}_{ct} \times \mathbb{1}(t \geq 2013) + \beta_3 \#Policy^{+}_{ct} + \gamma_c + \delta_t + \epsilon_{ct}, \tag{16}$$

where $\mathbf{X}_{ct}$ is a vector of city characteristics, including one of the two measures of city fiscal capacity discussed above, as well as GDP per capita, an indicator for local leadership turnover and an indicator for whether local leaders have personal ties to provincial officials. We also control for the total number of supportive industrial policies in a city-year $\#Policy^{+}_{ct}$, which captures the extensive margin of overall industrial policies in the city. In all specifications, we include the city and year fixed effects.

Columns (1) and (2) in Table 6 report the results. The coefficient estimates of the post-2013 interaction terms $\hat{\beta}_2$ imply that fiscally constrained cities are more likely to be top-down aligned in their choices of targeted sectors. Specifically, Column (1) shows that cities with higher land revenue, i.e. those with more abundant own revenue source, are less aligned with upper-level portfolios, while Column (2) shows that cities that are more dependent on intergovernmental transfers exhibit stronger top-down alignment in their choice of targeted industries.

In Columns (3) and (4), we report the estimation results where the dependent variable in the regression equation (16) is replaced by the RCA similarity measure $Similarity^{\text{RCA}}_{ct}$ as defined in Eq.(14). They show that the effect of the same fiscal variables on $Similarity^{\text{RCA}}_{ct}$ moves in the exact opposite direction after 2013 as on $Similarity^{p}_{ct}$, consistent with the view that after 2013, more fiscally constrained cities move away from the sectors with revealed local comparative advantages and toward top-down alignment with upper-level government's prioritized sectors. The coefficients on political turnover and connections are consistent with a signaling interpretation: the increase in top-down alignment modestly follows local leadership changes, while personal ties to upper-level officials attenuate the need to signal and are associated with lower top-down alignment.

[Table 6 about here]

Increasing top-down alignment also extends to *how* industrial policy is implemented. If local fiscal strain is an important driver for the increasing top down alignment, then the pass-through should be stronger when upper-level policies rely on tools that carry more direct fiscal resources. We test this by augmenting the city-level targeting specification with interactions between provincial targeting and the provincial tool mix:

$$\begin{aligned} Policy^{+}_{cst} = \exp\big[\beta_0 \, Policy^{p+}_{p(c)st} + \sum_k \beta_k \, Policy^{p+}_{p(c)st} \times Tool^{k}_{p(c)st} \\ + \sum_k \gamma_k \, Policy^{p+}_{p(c)st} \times Tool^{k}_{p(c)st} \times \mathbb{1}(t \geq 2013) + \delta_{cs} + \gamma_t\big] + \epsilon_{cst}. \end{aligned} \tag{17}$$

where $Tool^{k}_{p(c)st}$ indicates whether provincial-level industrial policies targeting industry s in year t deploy the tool category k. The coefficients β_k capture the effect of specific provincial-level policy tools on the top-down pass-through prior to 2013, and γ_k capture how the changes in the effect on pass-through associated with tool k changes after 2013.

Appendix Figure A5 reports the estimated $\hat{\beta}_k$ and $\hat{\gamma}_k$ for all policy tools. The post-2013 interaction coefficients γ_k are positive for most tool categories, but the largest increases are associated with fiscal and financial instruments. Provincial fiscal subsidies, investment-related policies, equity support, and tax incentives are tools containing the most direct fiscal support and are associated with the strongest effect on the increasing top-down alignment in the city-level choice of targeted sectors after 2013. Input-side policy tools – including labor policy and infrastructure investment – also exhibit a stronger post-2013 association with policy pass-through. In contrast, tools that are more closely tied to upgrading or longer-term objectives (e.g. environmental regulation, R&D and technology adoption, cluster promotion and some demand-side measures) show smaller post-2013 changes in their effect on top-down pass-through.

Local Political Promotion Incentive. A second channel operates through local politicians' *promotion incentives*. As the central government became more concrete and stronger in tone in its industrial policy directives, as seen in Figure 6, local compliance became easier to observe and potentially reward. To assess the role of local political promotion incentive in the increase in top-down alignment, we estimate a linear probability model for city leaders' promotion, allowing the coefficients to shift after 2013, namely:

$$\begin{aligned}\mathbb{1}(Promotion_{\ell,t+1}=1) = \alpha &+ \phi_1 \, Similarity^{p}_{c(\ell)t} + \phi_2 \, Similarity^{p}_{c(\ell)t} \times \mathbb{1}(t \geq 2013) \\ &+ \phi_3 \, Similarity^{\text{RCA}}_{c(\ell)t} + \phi_4 \, Similarity^{\text{RCA}}_{c(\ell)t} \times \mathbb{1}(t \geq 2013) + \mathbf{W}_{\ell t}\Gamma + \eta_c + \gamma_t, \quad (18)\end{aligned}$$

where ℓ is the mayor or party secretary of a city, $c(\ell)$ is the city of the leader ℓ, $\mathbb{1}(Promotion_{\ell,t+1} = 1)$ is a dummy variable that takes value 1 if the leader ℓ was promoted in year $t+1$, $\mathbf{W}_{\ell t}$ includes the characteristics of the leader and other controls, and η_c and γ_t respectively denote city and year fixed effects. We estimate the equation for mayors and party secretaries separately.

Table 7 reports the results. Before 2013, promotion was more strongly correlated with the alignment of the city's industrial policy portfolio with the local RCA ($Similarity^{\text{RCA}}_{c(\ell)t}$) than with the provincial government's portfolio ($Similarity^{p}_{c(\ell)t}$), consistent with an evaluation regime that placed greater weight on local economic performance. After 2013, however, the gradient pivots: for party secretaries, the post-2013 interaction on top-down alignment becomes positive and statistically significant, while for mayors, the alignment gradient also turns positive, though less precisely estimated.

[Table 7 about here]

Taken together, the evidence suggests that the post-2013 increase in top-down alignment in local governments' choices of targeted sectors operates not only through increasing concreteness and strength in central directives, but also through shifts in local fiscal incentives and promotion incentives that increase the local leaders' payoff from top-down compliance relative to aligning with local RCAs.

5.3 Potential Pitfalls of Chinese-Style Industrial Policies

Having shown in Section 5.2 that top-down alignment in China's industrial policy-making intensified after 2013–and that this shift was driven by stronger and broader central directives as well as changes in local fiscal and promotion incentives–we now study its consequences. We proceed in three steps. First, we compare the implications of *economic alignment* (targeting sectors that fit local RCAs) and *political alignment* (targeting sectors supported by upper-level governments) for firm entry and firm performance. Second, we examine policy diffusion by contrasting policy pioneers and followers. Third, we document how these micro patterns aggregate into changes in industrial policy portfolios and economic structure *across* cities, and examine their relationship with measures of intercity trade patterns.

Consequence of Increasing Political Alignment. We begin by relating local industrial policy to firm entry and outcomes, highlighting the differences between industrial policies that are consistent with economic versus political alignment. We consider the following outcomes: (i) entry at the city-industry-year level, measured by log of new registered capital; (ii) firm-level outcomes, including revenue-based TFP, log of firm profits, and log of average worker wages, constructed from administrative tax data; and (iii) measures of firms' product-market and labor-market wedges, namely product market markups and labor market wage markdowns, constructed following De Loecker and Warzynski (2012) and Brooks et al. (2021). The regression equation is specified as follows:

$$Y_{(f)cst} = \beta_1 Policy^{+}_{cst} + \beta_2 RCA^{p}_{cs(t-1)} + \beta_3 Policy^{+}_{cst} \times RCA^{p}_{cs(t-1)} + \beta_4 Policy^{p+}_{p(c)st} + \beta_5 Policy^{+}_{cst} \times Policy^{p+}_{p(c)st} + \delta_{cs} + \gamma_t + \epsilon_{(f)cst}, \quad (19)$$

where, depending on the outcome, the dependent variable is defined either at the city-industry-year level (Y_{cst}) or at the firm level (Y_{fcst}); the key explanatory variables of interest, $Policy^{+}_{cst}$, $RCA^{p}_{cs(t-1)}$, and $Policy^{p+}_{p(c)st}$, are the same as previously defined. All specifications include city-industry fixed effects δ_{cs} and year fixed effects γ_t.

Table 8 reports the results. Column (1) shows that local supportive policies toward a targeted sector are associated with higher firm entry, representing about 12% increase as measured by the newly registered capital; the positive and statistically significant coefficient estimates of the interaction terms, $\hat{\beta}_3$ and $\hat{\beta}_5$, suggests that the entry boosting effect is

stronger when the locally targeted sectors have higher local RCA and when they are also targeted by the provincial government's industrial policies.

Columns (2)-(4) show that local supportive policies are also positively associated with firm TFP, profits, and wages; however, these associations are substantially attenuated if the locally supported sector coincides with the provincial-level priority (i.e., when it is likely due to top-down political alignment), as reflected in the negative and statistically significant coefficient estimate of the interaction term $Policy^{+}_{cst} \times Policy^{p+}_{p(c)st}$.

Columns (5)-(6) show the corresponding results for product market markup and labor market wage markdown. Focusing on the interaction terms $Policy^{+}_{cst} \times RCA^{p}_{cs(t-1)}$ and $Policy^{+}_{cst} \times Policy^{p+}_{p(c)st}$, we find that economic alignment of the city industrial policy with the local RCA is positively associated with product market markup, but negatively associated with labor market wage markdown; in contrast, the political alignment of a city's industrial policy with provincial priorities is associated with lower product market markups and higher labor market wage markdowns. In conjunction with the increasing re-centralization in China's industrial policy landscape after 2013 which we previously documented, these findings provide an answer to the overcapacity and "involution" –increasing competitive pressure to lower product prices and mark down wages – in China.

[Table 8 about here]

Diminishing Returns to Policy Diffusion. If intensified competition and overcapacity are an important channel behind the weaker performance associated with re-centralization of China's industrial policy-making after 2013, then policies that are plausibly locally initiated should exhibit stronger returns than policies adopted later in follower waves. In Table A9, we show that later adopters are more likely to follow upper-level government's directives and less likely to target industries consistent with local comparative advantage. Beyond top-down directives, diffusion can also reflect learning: cities may imitate sectors and policy packages that appear successful elsewhere, particularly in fast-growing industries (e.g., high-tech and renewables). However, such imitation can reduce regional specialization if many cities converge on the same set of targeted industries.

To quantify how the effects of industrial policy vary with the diffusion sequence, we estimate the following equation:

$$Y_{(f)cst} = \beta_1 Policy^{+}_{cst} + \beta_2 Policy^{+}_{cst} \times Order_{cs} + \delta_{cs} + \gamma_t + \epsilon_{(f)cst}, \tag{20}$$

where $Order_{cs}$ is the adoption rank of city c for industry s, defined as the order in which the city first targets the industry;[19] the outcome variable $Y_{(f)cst}$ may be at the firm level (indicated by subscript f) or at the city level. The city-by-industry fixed effects δ_{cs} absorb time-invariant differences in a city's baseline propensity to target an industry (e.g.,

[19]We normalize the rank by the total number of cities (330), so that lower values correspond to earlier adoption (pioneers) and higher values to later adoption (followers).

persistent comparative advantage), and the year fixed effects, γ_t, capture year shocks common across all cities (or firms). The focus is on the interaction term $Policy^{+}_{cst} \times Order_{cs}$, which measures how the association between policy exposure and outcomes varies with the diffusion sequence.

Table 9 reports the results. Column (1) shows that supportive industrial policies in a city are associated with higher entry, and the entry response is even stronger for later adopters. Columns (2)-(6) show that supportive policies are associated with higher productivity, profits, and wages (and hence lower labor market wage markdowns) for early adopters, however, the coefficient estimates for the interaction term $Policy^{+}_{cst} \times Order_{cs}$ indicate that these associations attenuate as adoption becomes more widespread and can turn negative for later adopters, consistent with diminishing returns as later adopters face a more crowded market and possibly deviate more from their local comparative advantages.

[Table 9 about here]

We would like to argue that one of the reasons for the diminishing return to policy diffusion is that later adopters may rely more on policy tool bundles that mirror upper-level templates with less local adaptation. To study the choice of local governments in their bundles of policy tools, we construct a tool vector for each city-industry-year, where the k-th element equals one if the policy tool k is used to support industry s in the city c in year t. We then compute the cosine similarity between the tool vector of a city and the corresponding provincial tool vector for the same industry-year. Figure 8 presents the binscatter plot of this tool-similarity measure against $Order_{cs}$, after absorbing the city, industry, and year fixed effects. It shows that tool similarities with the provincial government (Panel A) and with the central government (Panel B) both increase monotonically with the city's adoption order, indicating that later adopters are more likely to replicate upper-level tool bundles and exhibit less local adaptation.[20]

[Figure 8 about here]

Policy Persistence versus Self-correction. Having shown that policy diffusion driven by top-down alignment is associated with excessive entry, thinner profit margins (lower product price markup), and higher labor market wage markdowns, we ask whether local governments learn from these negative market signals and discontinue their supportive policies. Specifically, we study the dynamic link between market outcomes and the continuation of local targeting at the city–industry level. Intuitively, a self-correction mechanism implies that, conditional on having targeted a sector in the recent past, local governments are less likely to continue when entry becomes crowded, product-market margins compress,

[20]For Figure 8, we use the raw order without normalizing by the number of prefectural level cities in China.

or labor market wage markdowns rise. Conversely, top-down alignment incentives may imply weak or even opposite sensitivity to these signals, leading to policy persistence despite negative signals.

We measure persistence as the probability that an industry remains targeted in city c in year t conditional on past targeting. We focus on the sample of policy-targeted city-industry-year cells and estimate a PPML specification on whether the industry is still being supported 1-3 years later:

$$Policy^{+}_{cs(t+k)} = \exp\Big[\beta_k \Delta_k X_{cst} + \lambda_1 Policy^{p+}_{p(c)st} + \lambda_2 Policy^{p+}_{p(c)s(t+k)} + \eta_{cs} + \gamma_t\Big] + \epsilon_{cst}, \quad (21)$$

where $\Delta_k X_{cst}$ denotes the k-year change in market outcomes for city c and industry s from year t to $t+k$, including log new registered capital (from firm registration data), and product market markups and labor market wage markdowns (aggregated to the city-industry-year level, estimated from the administrative tax records data). The city-industry fixed effects η_{cs} absorb time-invariant heterogeneity across cities by industries, and the year fixed effects γ_t absorb time-dependent shocks common to all cities and industries. We also control for contemporaneous and future provincial targeting, $Policy^{p+}_{p(c)st}$ and $Policy^{p+}_{p(c)s(t+k)}$, to account for the potential role of upper-level policies in local government's choices of which sector to target. We take $k \in \{1, 2, 3\}$ to capture one to three years of short-run dynamics. The coefficients have a direct interpretation. If the local governments self correct in their industrial policy targeting, we expect that $\beta_k < 0$ when X is the change in log new registered capital (i.e., more entry reduces continuation of supportive industrial policies), $\beta_k > 0$ when X is the change in product market markups (i.e., product market margin compression reduces continuation), and $\beta_k < 0$ when X is the change in labor market wage markdowns (i.e., stronger downward wage pressures in the sector reduces continuation of supportive industrial policies).

Table 10 reports the results. We find little evidence of self-correction. To the contrary, industrial policy is more likely to continue when new firm entries rise, product market markups fall, and labor market wage markdowns intensify. In addition, provincial targeting significantly increases local persistence in their industrial policy support.

[Table 10 about here]

Local Protectionism. The macro consequences follow directly from the mechanism that we have established. Stronger and more dispersed central guidance after 2013, combined with heightened top-down alignment, broadens the set of sectors each locality targets (a decline in local policy concentration) while also pushing local portfolios toward a common template (an increase in cross-city policy similarity). This combination may result in more dispersed industrial structures within cities, yet more similar structures across cities, eroding intercity complementarities in their industrial structures, which in turn weakens the incentives for cross-city trade. This exacerbates local protectionism, which has been a

persistent concern in China, driven by fiscal and political motives (Young, 2000; Fang, Li, and Wu, 2025).

To test the above hypothesis, we first measure industrial policy similarity across cities in the choice of policy-targeted industries: for each pair of cities c and c' within the same province, we compute the cosine similarity of their policy vectors $\mathbf{P}_{ct}$ and $\mathbf{P}_{c't}$ analogous to Eq.(13). We then summarize a city's policy similarity in year t as the average of its pairwise cosine similarities with all other cities in the same province.

To relate policy similarity to trade patterns, we use firm-to-firm VAT transaction data matched to firm registration records, which allows us to assign firms to cities. We aggregate transactions to the city-pair level and construct two complementary measures of cross-city transactions. At the city level, we compute the share of intra-city transactions in a city's total transactions in the province. Panel (A) of Figure 9 shows that this measure of home-bias is positively correlated with the city's policy-similarity index within the province: cities whose policy portfolios more closely resemble those of other cities in the province conduct a higher share of transactions within their own borders. At the city-pair level, we compute the share of bilateral intercity transactions in the combined total transaction volume of the two cities and relate it to the pairwise policy similarity. Panel (B) shows a negative correlation: city pairs with more similar policies trade less with each other. Together, these patterns are consistent with weaker intercity complementarities when industrial policy portfolios become more similar across cities.

[Figure 9 about here]

6 Conclusion

In this paper, we construct a multi-dimensional and granular measure of China's industrial policy from a near-universe of official government documents issued between 2000 and 2022. Using an LLM-based extraction pipeline, we classify industrial policy documents issued by national, provincial, and importantly, prefectural city governments, and recover key attributes, including policy tone, targeted industries, and a detailed set of implementation tools. The resulting database provides an empirical foundation for studying not only which sectors are targeted by each city-level government, how industrial policies propagate through the Chinese political hierarchy, but also how policies are implemented over the life-cycle of targeted industries.

Using these data, we first examine the economic and political economic determinants of local governments' choices of targeted industries. We show that local governments disproportionately target industries that align with local economic fundamentals, as proxied by capital-based revealed comparative advantage and/or absolute advantage, as well as sectors that are central in local production networks and connected through upstream and downstream linkages with other supported sectors; at the same time, we find that city-level

targeting exhibits substantial top-down pass-through from national and provincial governments' choices of targeted industries. Interestingly, we find that the extent of top-down pass-through of targeted industries varies systematically with local conditions: it is weaker in richer cities, stronger where within-province promotion competition is more intense, and attenuated when local leaders have personal ties to upper-level officials. These patterns indicate that the choice of the local portfolio of targeted sectors reflects both the local economic fundamentals and the career incentives provided by China's hierarchical political structure.

We further show that the balance between the local economic fundamentals and top-down alignment shifts markedly after 2013, specifically, the importance of top-down alignment rises while that of the fit with local revealed comparative advantage declines. We link this shift in balance around 2013 to both changes in central policy directives and changes in local governments' incentives. Specifically, we show that the central government's policy directives become more concrete and stronger in tone; moreover, the sectoral coverage of central government's industrial policies, as measured by the Herfindahl-Hirschman index of the central government's policy portfolio, becomes broader. On the local side, we show that the increasing top-down alignment of the local industrial policy portfolio is amplified by the city's fiscal dependence and the local officials' promotion incentives: fiscally constrained cities become more aligned with upper-level portfolios because supporting the industries supported by upper-level governments results in more fiscal transfers to the city; and the association between local officials' promotions and top-down alignment increases after 2013. In addition, we show that re-centralization after 2013 also extends from which sectors are targeted to how they are supported: consistent with the aforementioned fiscal channel, we show that pass-through strengthens most for fiscally salient policy tools by provincial governments.

The post-2013 shift toward re-centralization in China's industrial policy landscape has important micro and macro consequences. Distinguishing local industrial policy driven by economic alignment from that driven by top-down political alignment, we find that both types of local targeting are associated with higher firm entry (as measured by newly registered capital), but their implications for firm outcomes, such as TFP, profit, and "wedges" such as product market markup and labor market wage markdown, diverge. Policies that are more closely aligned with local comparative advantage are associated with more favorable firm outcomes and "wedges," (i.e. higher product-market markups and lower labor market wage markdowns) whereas top-down aligned targeting is associated with weaker performance, compressed product-market markups, and higher labor market wage markdowns. These patterns are reinforced along the chain of diffusion: follower cities' supportive industrial policies exhibit stronger firm entry responses but diminishing, and in some cases negative, associations with firm TFP, profits, and wages as markets become more crowded; also, later adopters are increasingly likely to rely on more templated policy tool bundles that are similar to those of the upper-level government with much less local adaptation.

Furthermore, we find little evidence of local self-correction: conditional on a sector being supported by a city's industrial policy, continuation of the supportive policy is more likely when market signals such as product market markups and labor market wage markdowns deteriorate as long as it had spurred new firm entry, and upper-level supportive policies for the sector further increase local policy persistence. Aggregating across cities, these results imply that after 2013, China's industrial policy landscape and the resulting economic structure exhibited high similarity across cities, and this, in turn, led to enhanced local protection bias and weaker intercity exchange in economic transactions.

Substantively, the Chinese evidence highlights a general political-economy tension in a multi-tiered industrial policy landscape: stronger coordination can scale priorities and accelerate diffusion, but when top-down compliance incentives are strengthened through fiscal dependence and career concerns, top-down political alignment may attenuate local adaptation and contribute to wasteful duplication and exacerbate local protectionism.

Our database opens several avenues for future research. First, while we document systematic heterogeneity across tools and contexts, quantifying welfare consequences and isolating mechanisms requires structural and general-equilibrium frameworks that incorporate selection, congestion, and network spillovers. Second, understanding the persistence of inefficient targeting at the local level calls for a deeper analysis of the political and institutional frictions that impede exit, restructuring, and reallocation of "zombie" firms including barriers to mergers, acquisitions, and bankruptcy. More broadly, the measurement approach developed here can be applied to other settings to study how industrial policy design and implementation vary across jurisdictions and over time, and how those choices shape firm dynamics and market structure.

References

Aghion, Philippe, Jing Cai, Mathias Dewatripont, Luosha Du, Ann Harrison, and Patrick Legros. 2015. "Industrial Policy and Competition." *American Economic Journal: Macroeconomics* 7 (4):1–32.

Bai, John Jianqiu, Nicole M Boyson, Yi Cao, Miao Liu, and Chi Wan. 2023. "Executives vs. Chatbots: Unmasking Insights through Human-AI Differences in Earnings Conference Q&A." *Available at SSRN 4480056* .

Balassa, Bela. 1965. "Trade Liberalisation and "Revealed" Comparative Advantage." *The Manchester School* 33 (2):99–123.

Baldwin, Richard and Paul R Krugman. 1986. "Market Access and International Competition: A Simulation Study of 16k Random Access Memories." *NBER Working Paper No. 1936* .

Baldwin, Robert E. 1969. "The Case against Infant-Industry Tariff Protection." *Journal of Political Economy* 77 (3):295–305.

Barwick, Panle Jia, Myrto Kalouptsidi, and Nahim Bin Zahur. 2025. "Industrial Policy Implementation: Empirical Evidence from China's Shipbuilding Industry." *Review of Economic Studies* 92 (6):3611–3648.

Bo, Shiyu. 2020. "Centralization and Regional Development: Evidence from a Political Hierarchy Reform to Create Cities in China." *Journal of Urban Economics* 115:103182.

Branstetter, Lee G and Guangwei Li. 2023. "The Challenges of Chinese Industrial Policy." *NBER Working Paper No. 14871* .

Branstetter, Lee G, Guangwei Li, and Mengjia Ren. 2023. "Picking Winners? Government Subsidies and Firm Productivity in China." *Journal of Comparative Economics* 51 (4):1186–1199.

Brooks, Wyatt J, Joseph P Kaboski, Yao Amber Li, and Wei Qian. 2021. "Exploitation of Labor? Classical Monopsony Power and Labor's Share." *Journal of Development Economics* 150:102627.

Bybee, J Leland. 2023. "The Ghost in the Machine: Generating Beliefs with Large Language Models." *arXiv preprint arXiv:2305.02823* .

Calcagno, Alfredo, Sebastian Dullien, Alejandro Márquez-Velázquez, Nicolas Maystre, and Jan Priewe. 2015. *Rethinking Development Strategies After the Financial Crisis*. United Nations.

Cen, Xiao, Vyacheslav Fos, and Wei Jiang. 2024. "How Do US Firms Withstand Foreign Industrial Policies?" *NBER Working Paper No. 32411* .

Chen, Donghua, Oliver Zhen Li, and Fu Xin. 2017. "Five-Year Plans, China Finance and Their Consequences." *China Journal of Accounting Research* 10 (3):189–230.

Chen, Yi, Hanming Fang, Yi Zhao, and Zibo Zhao. 2024. "Recovering Overlooked Information in Categorical Variables with LLMs: An Application to Labor Market Mismatch." *NBER Working Paper No. 32327* .

Choi, Jaedo and Andrei A Levchenko. 2025. "The Long-Term Effects of Industrial Policy." *Journal of Monetary Economics* 152:103779.

De Loecker, Jan and Frederic Warzynski. 2012. "Markups and Firm-Level Export Status." *American Economic Review* 102 (6):2437–2471.

DiPippo, Gerard, Ilaria Mazzocco, Scott Kennedy, and Matthew P Goodman. 2022. *Red Ink: Estimating Chinese Industrial Policy Spending in Comparative Perspective*. Center for Strategic & International Studies (CSIS) Report. May.

Eisfeldt, Andrea L, Gregor Schubert, and Miao Ben Zhang. 2023. "Generative AI and Firm Values." *NBER Working Paper No. 31222* .

Evenett, Simon, Adam Jakubik, Fernando Martín, and Michele Ruta. 2024. "The Return of Industrial Policy in Data." *The World Economy* 47 (7):2762–2788.

Fang, Hanming, Quanlin Gu, Wei Xiong, and Li-An Zhou. 2016. "Demystifying the Chinese Housing Boom." *NBER Macroeconomics Annual* 30 (1):105–166.

Fang, Hanming, Ming Li, and Zenan Wu. 2025. "Tournament-Style Political Competition and Local Protectionism: Theory and Evidence from China." *Journal of Public Economics* 248:105421.

Fang, Hanming and Rongjie Zhang. 2025. "Corruption Stereotype and the Unintended Consequences of an Anti-Corruption Campaign: Evidence from the Real Estate Sector in China." *Journal of Public Economics* 249:105474.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy. 2019. "Text as Data." *Journal of Economic Literature* 57 (3):535–574.

Goldberg, Pinelopi K, Réka Juhász, Nathan J Lane, Giulia Lo Forte, and Jeff Thurk. 2024. "Industrial Policy in the Global Semiconductor Sector." *NBER Working Paper No. 32651* .

Goldstein, Itay, Chester S Spatt, and Mao Ye. 2021. "Big Data in Finance." *Review of Financial Studies* 34 (7):3213–3225.

Greenwald, Bruce and Joseph E Stiglitz. 2006. "Helping Infant Economies Grow: Foundations of Trade Policies for Developing Countries." *American Economic Review* 96 (2):141–146.

Hansen, Jørgen Drud, Camilla Jensen, and Erik Strøjer Madsen. 2003. "The Establishment of the Danish Windmill Industry—Was It Worthwhile?" *Review of World Economics* 139:324–347.

Harrison, Ann and Andrés Rodríguez-Clare. 2010. "Trade, Foreign Investment, and Industrial Policy for Developing Countries." *Handbook of Development Economics* 5:4039–4214.

Head, Keith. 1994. "Infant Industry Protection in the Steel Rail Industry." *Journal of International Economics* 37 (3-4):141–165.

Imbert, Clement, Marlon Seror, Yifan Zhang, and Yanos Zylberberg. 2022. "Migrants and Firms: Evidence from China." *American Economic Review* 112 (6):1885–1914.

Irwin, Douglas A. 2000. "Could the United States Iron Industry Have Survived Free Trade after the Civil War?" *Explorations in Economic History* 37 (3):278–299.

Itskhoki, Oleg and Benjamin Moll. 2019. "Optimal Development Policies with Financial Frictions." *Econometrica* 87 (1):139–173.

Jha, Manish, Jialin Qian, Michael Weber, and Baozhong Yang. 2024. "ChatGPT and Corporate Policies." *NBER Working Paper No. 32161* .

Juhász, Réka. 2018. "Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade." *American Economic Review* 108 (11):3339–3376.

Juhász, Réka, Nathan Lane, Emily Oehlsen, and Veronica C Pérez. 2025. "Measuring Industrial Policy: A Text-Based Approach." *NBER Working Paper No. 33895* .

Juhász, Réka, Nathan Lane, and Dani Rodrik. 2024. "The New Economics of Industrial Policy." *Annual Review of Economics* 16 (1):213–242.

Kim, Alex, Maximilian Muhn, and Valeri V Nikolaev. 2024. "Bloated Disclosures: Can ChatGPT Help Investors Process Information?" *Chicago Booth Research Paper 23-07* .

Korinek, Anton. 2023. "Generative AI for Economic Research: Use Cases and Implications for Economists." *Journal of Economic Literature* 61 (4):1281–1317.

Krueger, Anne O. 1990. "Government Failures in Development." *Journal of Economic Perspectives* 4 (3):9–23.

Krugman, Paul. 1992. *Geography and Trade*. MIT press.

Lane, Nathan. 2020. "The New Empirics of Industrial Policy." *Journal of Industry, Competition and Trade* 20:209–234.

———. 2025. "Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea." *Quarterly Journal of Economics* 140 (3):1683–1741.

Lee, Sangkuk. 2017. "An Institutional Analysis of Xi Jinping's Centralization of Power." *Journal of Contemporary China* 26 (105):325–336.

Li, Edward Xuejun, Zhiyuan Tu, and Dexin Zhou. 2024. "The Promise and Peril of Generative AI: Evidence from ChatGPT as Sell-Side Analysts." *Available at SSRN 4480947* .

Li, Guangwei and Lee G Branstetter. 2024. "Does "Made in China 2025" Work for China? Evidence from Chinese Listed Firms." *Research Policy* 53 (6):105009.

Li, Junyou, Qin Zhang, Yangbin Yu, Qiang Fu, and Deheng Ye. 2024. "More Agents Is All You Need." *arXiv preprint: arXiv:2402.05120* .

Li, Kai, Feng Mai, Rui Shen, Chelsea Yang, and Tengfei Zhang. 2025. "Dissecting Corporate Culture Using Generative AI–Insights from Analyst Reports." *Available at SSRN 4558295* .

Lin, Justin and Ha-Joon Chang. 2009. "Should Industrial Policy in Developing Countries Conform to Comparative Advantage or Defy It? A Debate Between Justin Lin and Ha-joon Chang." *Development Policy Review* 27 (5):483–502.

Lin, Justin Yifu. 2015. *The Quest for Prosperity: How Developing Economies Can Take Off*. Princeton University Press.

Lin, Justin Yifu, Celestin Monga, and Joseph Stiglitz. 2013. "The Rejuvenation of Industrial Policy." *World Bank Policy Research Working Paper 6628* .

Liu, Ernest. 2019. "Industrial Policies in Production Networks." *Quarterly Journal of Economics* 134 (4):1883–1948.

Lopez-Lira, Alejandro and Yuehua Tang. 2023. "Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models." *arXiv preprint arXiv:2304.07619* .

Lü, Xiaobo and Pierre F Landry. 2014. "Show Me the Money: Interjurisdiction Political Competition and Fiscal Extraction in China." *American Political Science Review* 108 (3):706–722.

Luo, Kaicheng, Shaoda Wang, and David Y Yang. 2025. "Laboratories of Autocracy: Landscape of Central–Local Dynamics in China's Policy Universe." *NBER Working Paper No. 34219* .

Luzio, Eduardo and Shane Greenstein. 1995. "Measuring the Performance of a Protected Infant Industry: the Case of Brazilian Microcomputers." *Review of Economics and Statistics* 77 (4):622–633.

Madaan, Aman, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang et al. 2023. "Self-Refine: Iterative Refinement with Self-Feedback." *Advances in Neural Information Processing Systems* 36:46534–46594.

Manelici, Isabela and Smaranda Pantea. 2021. "Industrial Policy at Work: Evidence from Romania's Income Tax Break for Workers in It." *European Economic Review* 133:103674.

Melitz, Marc J. 2005. "When and How Should Infant Industries Be Protected?" *Journal of International Economics* 66 (1):177–196.

Naughton, Barry. 2021. *The Rise of China's Industrial Policy, 1978 to 2020*. Universidad Nacional Autónomica de México, Facultad de Economía México.

Naughton, Barry, Siwen Xiao, and Yaosheng Xu. 2023. "The Trajectory of China's Industrial Policies." *IGCC Working Paper No 6* .

Pack, Howard. 2000. "Industrial Policy: Growth Elixir or Poison?" *The World Bank Research Observer* 15 (1):47–67.

Redding, Stephen. 1999. "Dynamic Comparative Advantage and the Welfare Effects of Trade." *Oxford Economic Papers* 51 (1):15–39.

Rodriguez-Clare, Andres. 2007. "Clusters and Comparative Advantage: Implications for Industrial Policy." *Journal of Development Economics* 82 (1):43–57.

Rodrik, Dani. 2004. "Industrial Policy for the Twenty-First Century." *CEPR Discussion Papers 4767* .

———. 2009. "Industrial Policy: Don't Ask Why, Ask How." *Middle East Development Journal* 1 (1):1–29.

Shinn, Noah, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. "Reflexion: Language Agents with Verbal Reinforcement Learning." *Advances in Neural Information Processing Systems* 36.

Silva, JMC Santos and Silvana Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88 (4):641–658.

Silva, JMC Santos and Rainer Winkelmann. 2024. "Misspecified Exponential Regressions: Estimation, Interpretation, and Average Marginal Effects." *Review of Economics and Statistics* :1–25.

Sinclair, Andrew J. and Chuyi Zhang. 2025. "Discounting Industrial Policy." *Available at SSRN 5147033* .

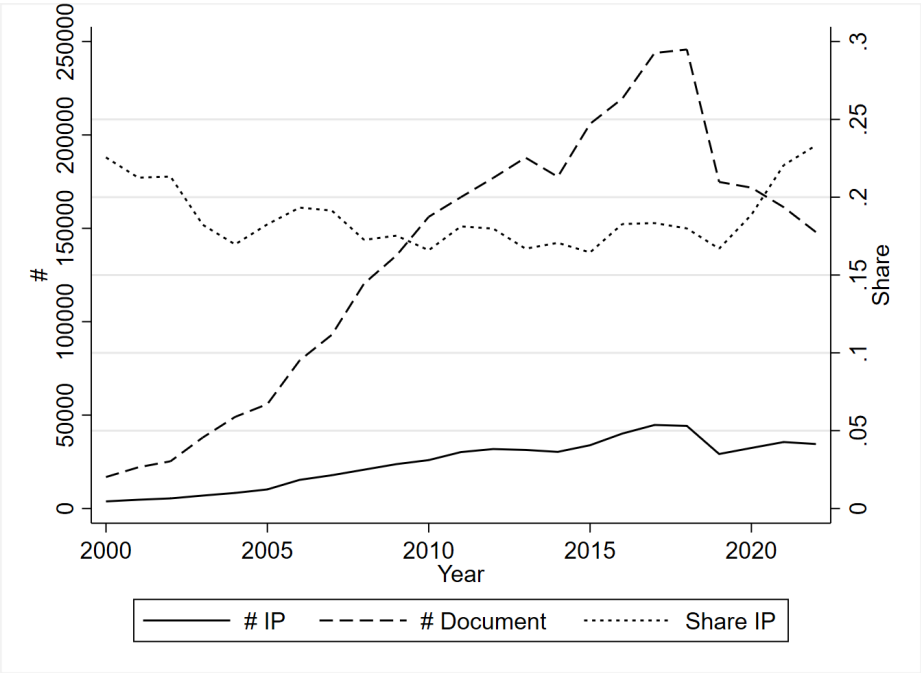
Wei, Shang-Jin, Jianhuan Xu, Ge Yin, and Xiaobo Zhang. 2023. "Mild Government Failure." *NBER Working Paper No. 31178* .

Xu, Chenggang. 2011. "The Fundamental Institutions of China's Reforms and Development." *Journal of Economic Literature* 49 (4):1076–1151.

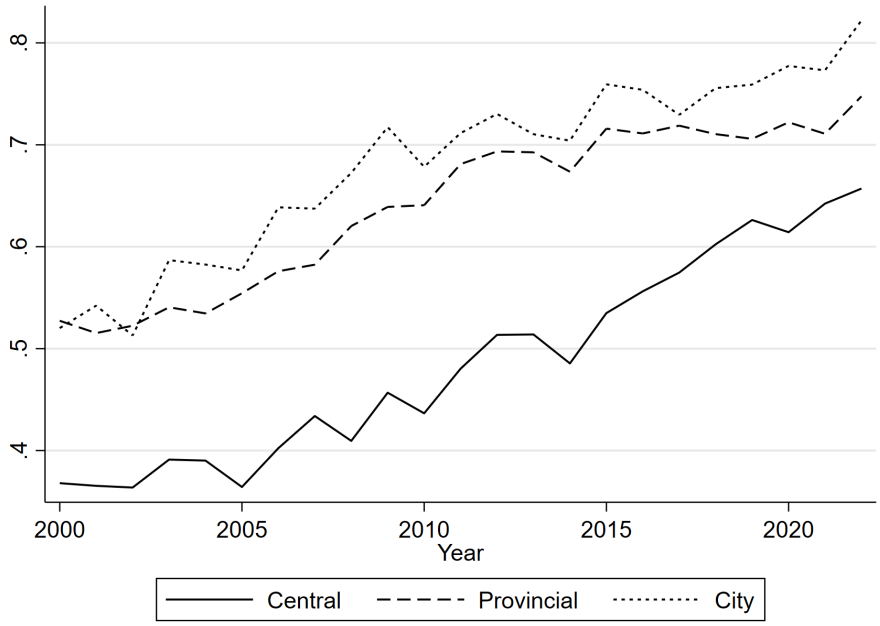
Young, Alwyn. 2000. "The Razor's Edge: Distortions and Incremental Reform in the People's Republic of China." *Quarterly Journal of Economics* 115 (4):1091–1135.

Zhou, Hui, Junqiang Liu, Jiang He, and Jianxin Cheng. 2021. "Conditional Justice: Evaluating the Judicial Centralization Reform in China." *Journal of Contemporary China* 30 (129):434–450.

Figure 1: Evolution of the Industrial Policies

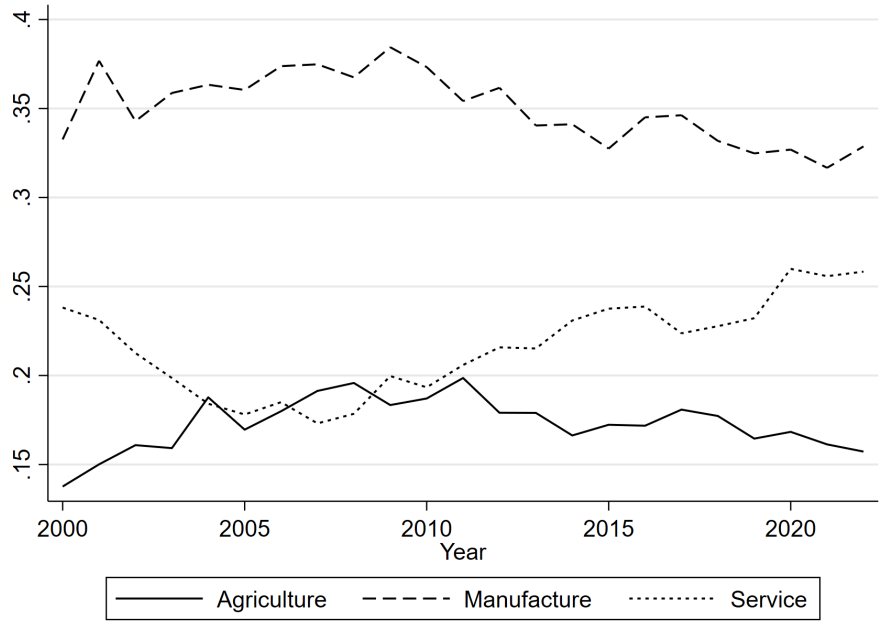


(A) Overall

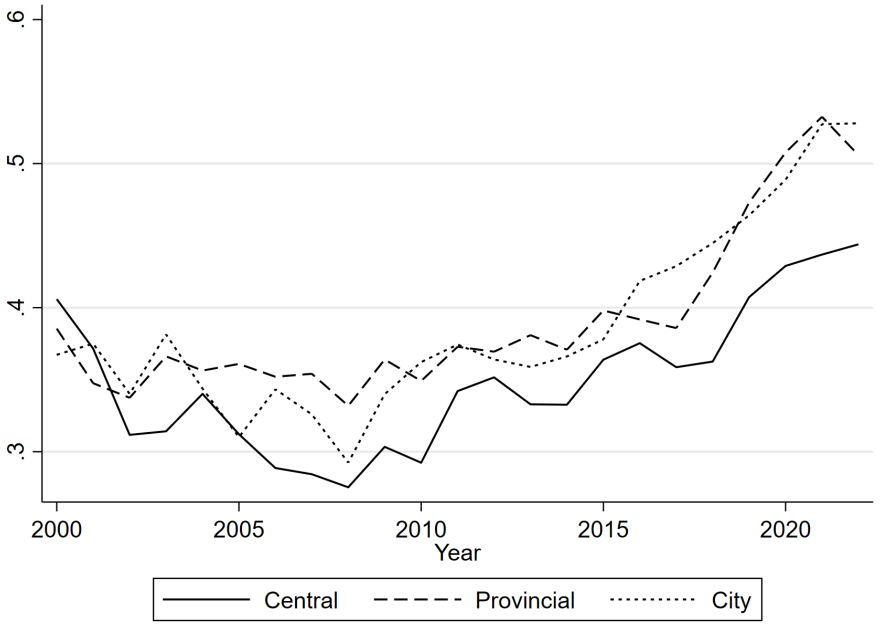


(B) Fraction of Supportive Policies

Figure 2: Rising Importance of High-skill Manufacturing Industries

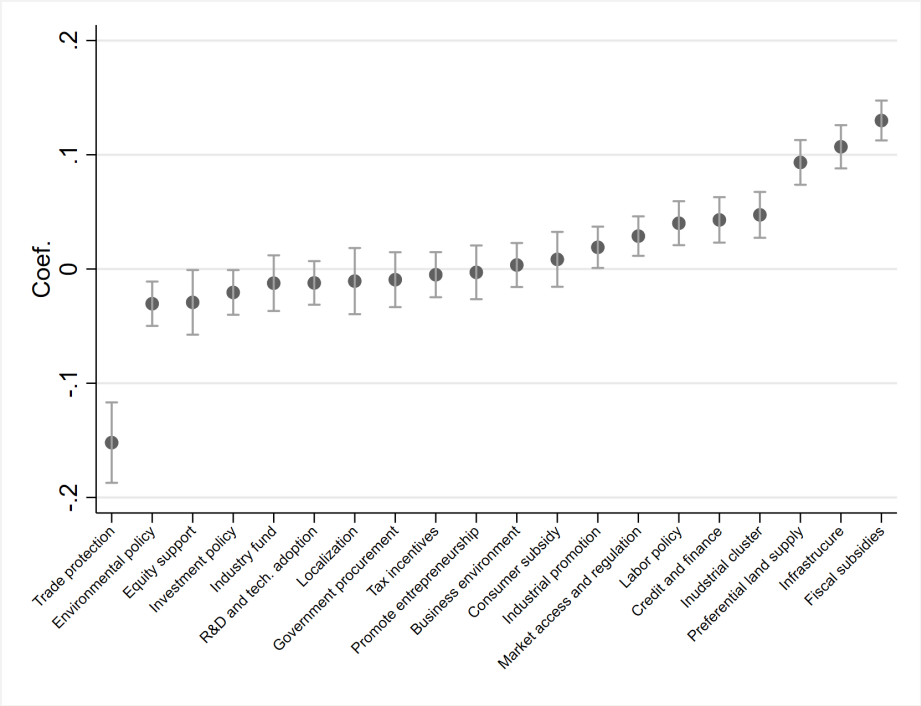


(A) Overall Shares Targeting Agriculture, Manufacturing and Service Sectors

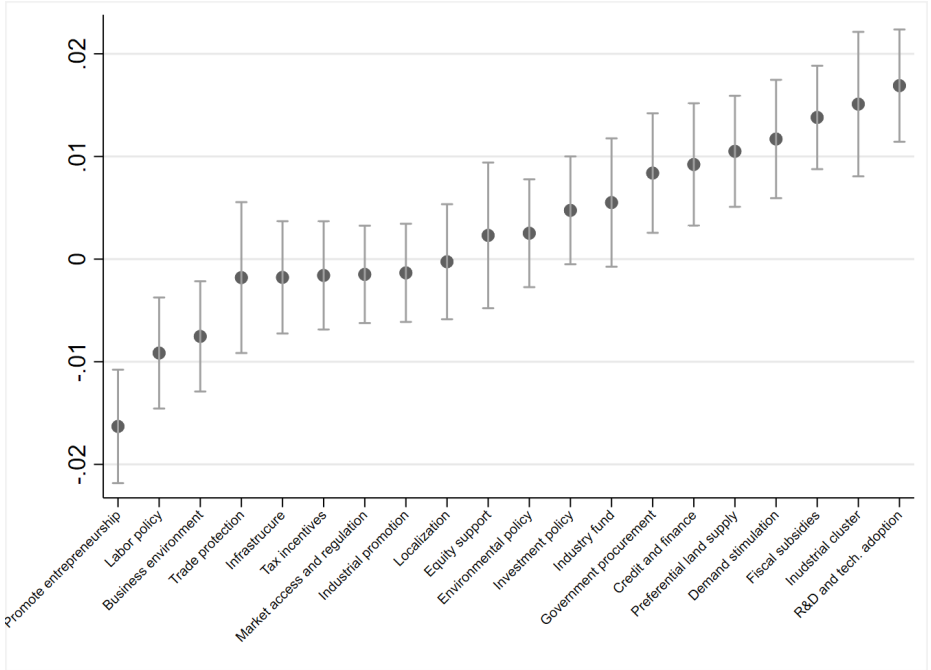


(B) Share of High-Skill Manufacturing within the Manufacturing Sector

Figure 3: Effects of Industrial Policy Tools on New Firm Entry



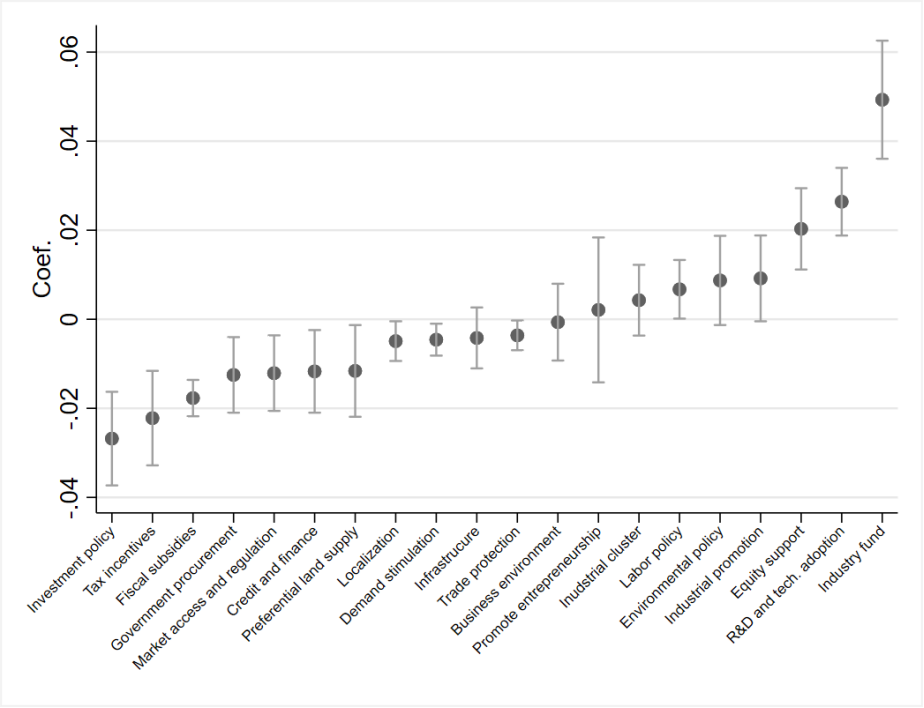
(A) Entry



(B) TFP

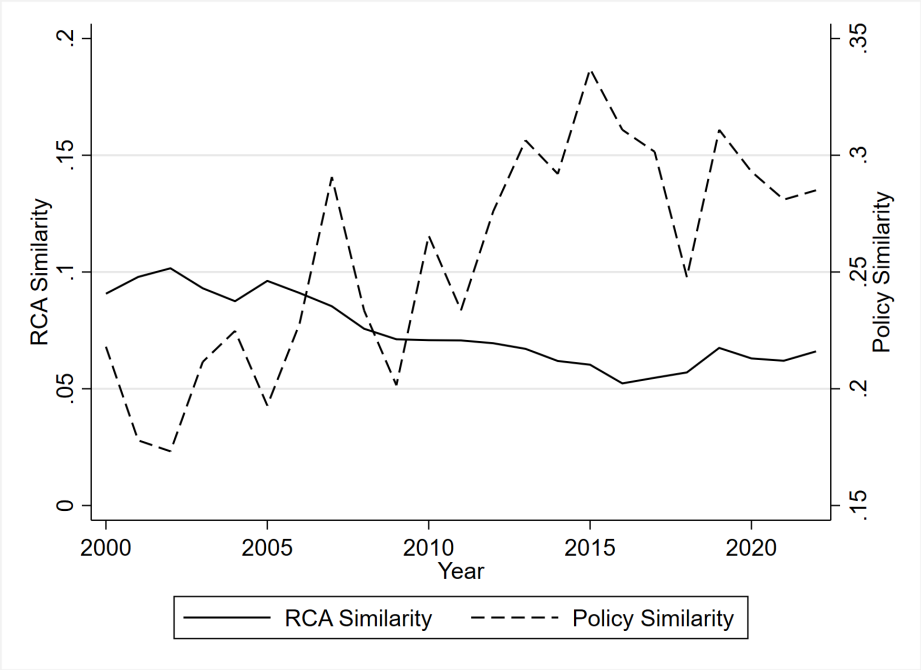
Notes: This figure plots the coefficients and the corresponding 95% confidence intervals for each industrial policy tool by estimating Equation (3). Unit of observation is city-industry-year for Panel (A) and firm-year for Panel (B). Standard errors are clustered at the city-by-industry level.

Figure 4: Within-industry Change in Policy Tool

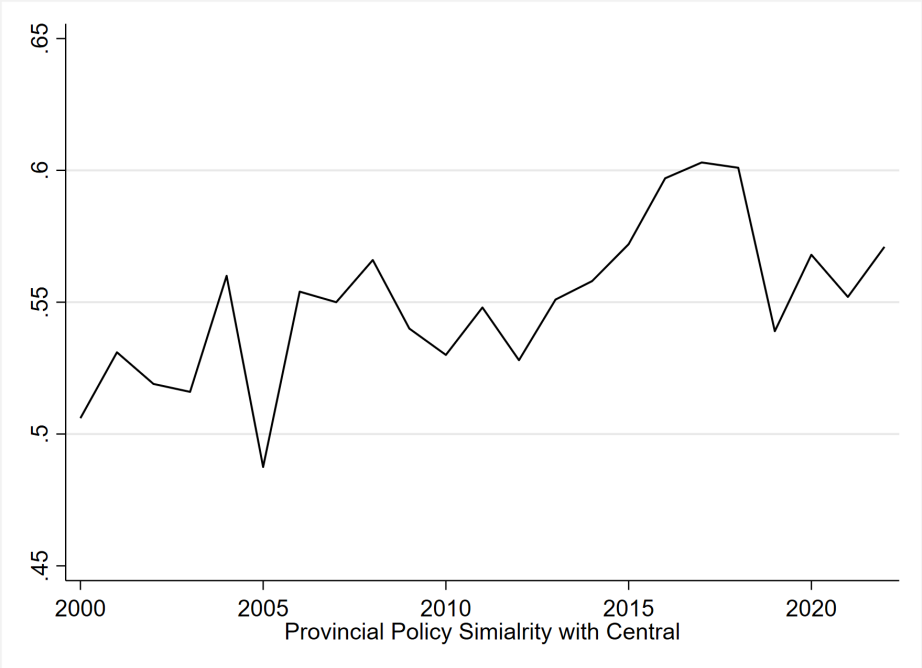


Notes: This figure plots the estimated coefficients for β_k for each industrial policy tool by estimating Equation (4), where a positive coefficient indicates an increased likelihood of tool utilization as the industry matures. Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level.

Figure 5: 2013 as a Turning Point: Evolution of Similarity Index



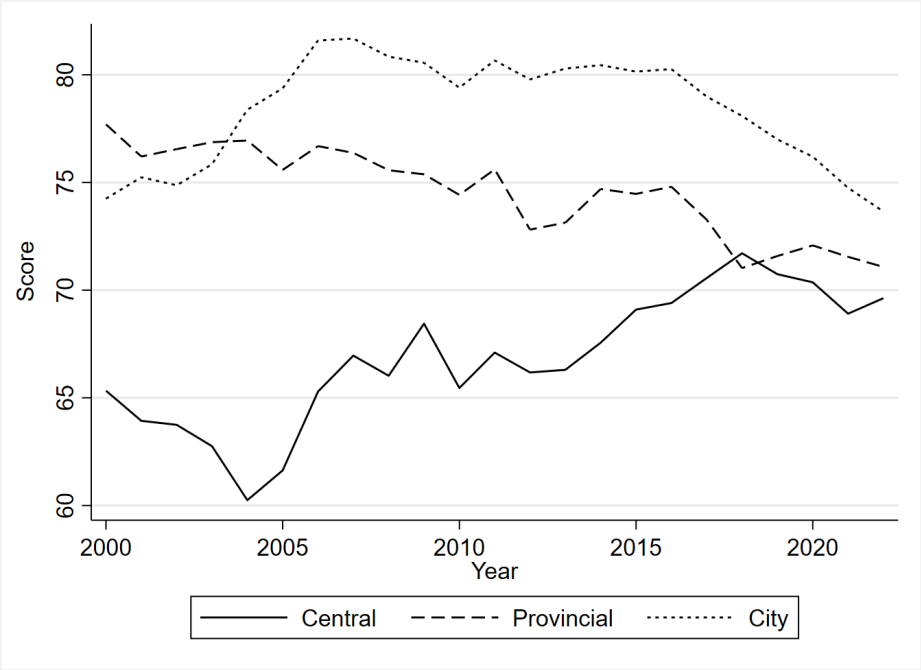
(A) City Level



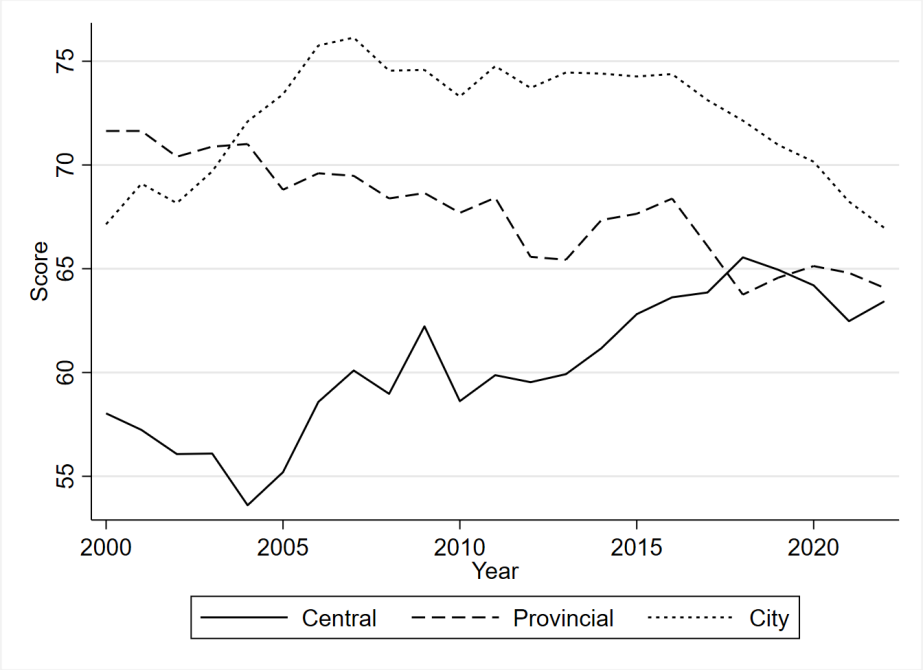
(B) Province Level

Notes: This figure plots the evolution of the similarity measures over time, with the 95% confidence interval indicating the cross-city distribution within each year.

Figure 6: Central Government Turns More Assertive in Policy Measures



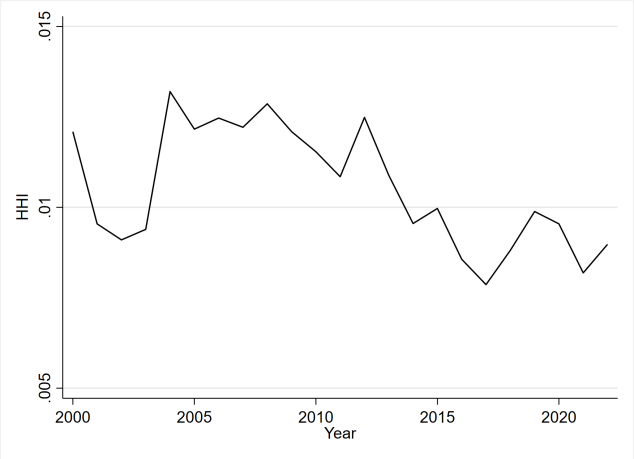
(A) Policy Concreteness



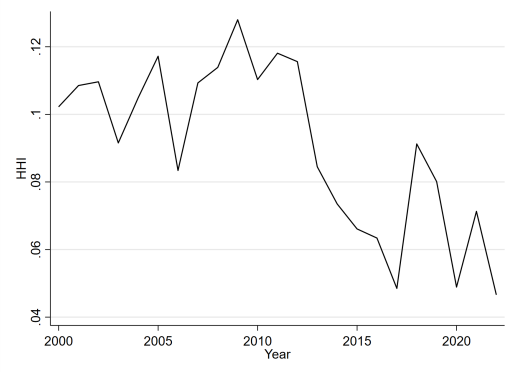
(B) Policy Strength

Notes: This figure plots the average score for policy concreteness (Panel A) and policy strength (Panel B) over time, by government level.

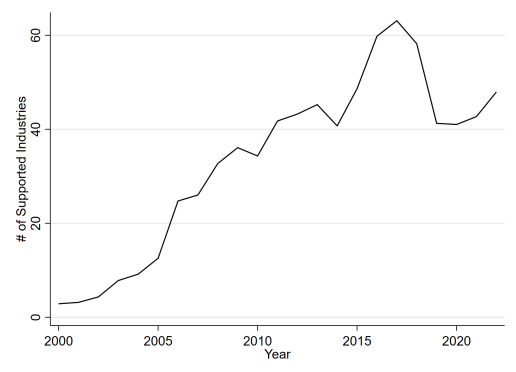
Figure 7: Policies are Less Concentrated Across Industries



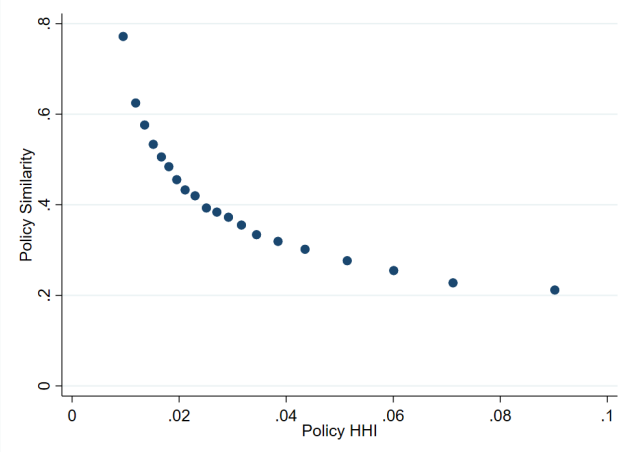
(A) Central Policy HHI



(B) City Policy HHI



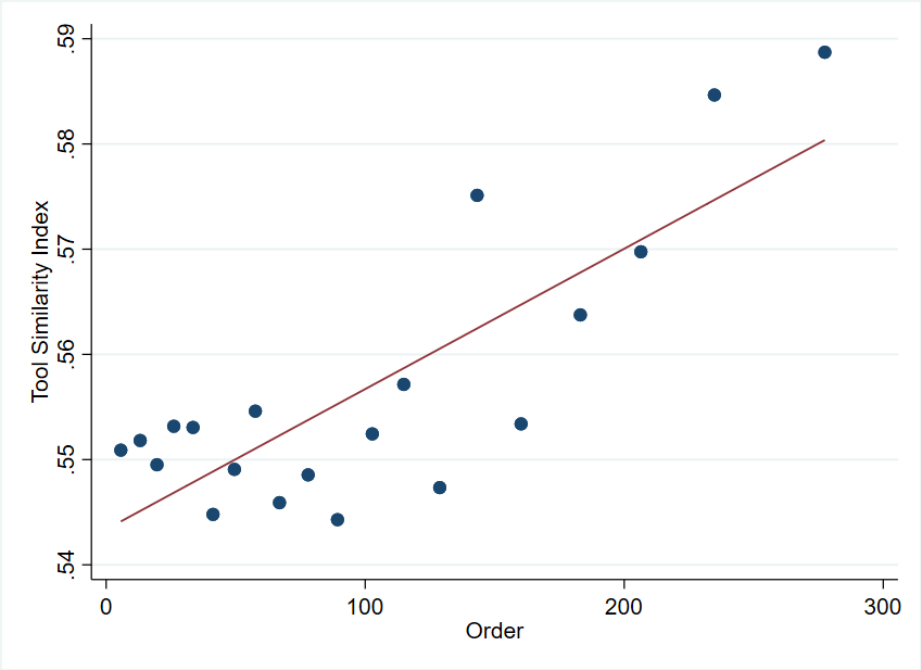
(C) City # of Supported Industries



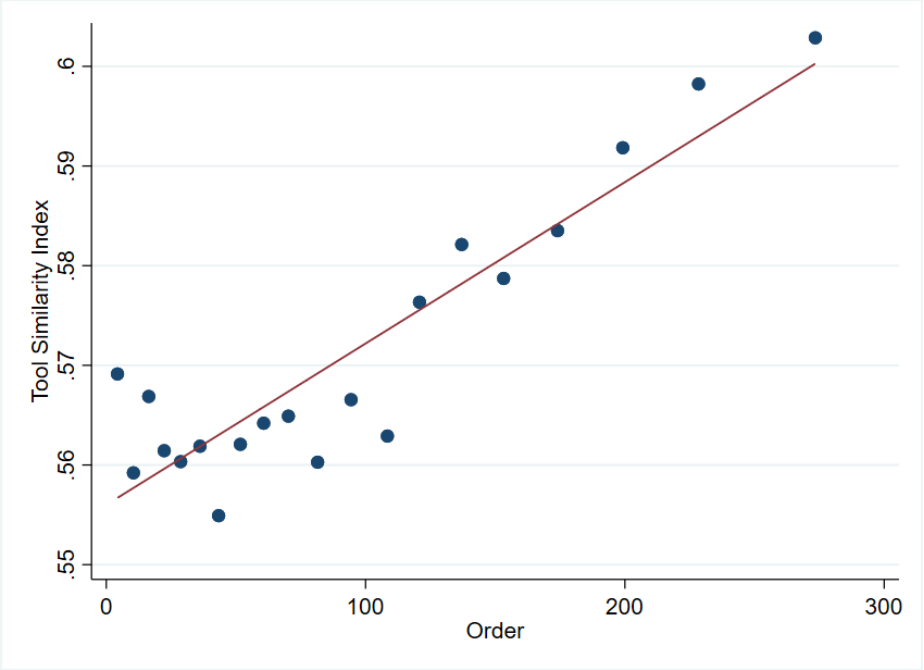
(D) Policy HHI and Policy Similarity

Notes: This figure plots the policy coverage (number of supported industries, and HHI across industries) over time. Panel (A) is for the central policy, Panel (B) for city-level policy, Panel (C) reports the number of industries supported by city-level policy, and Panel (D) is the binscatterplot showing the correlation between city level policy HHI and policy similarity with the provincial policy.

Figure 8: Policy Diffusion and Policy Tool



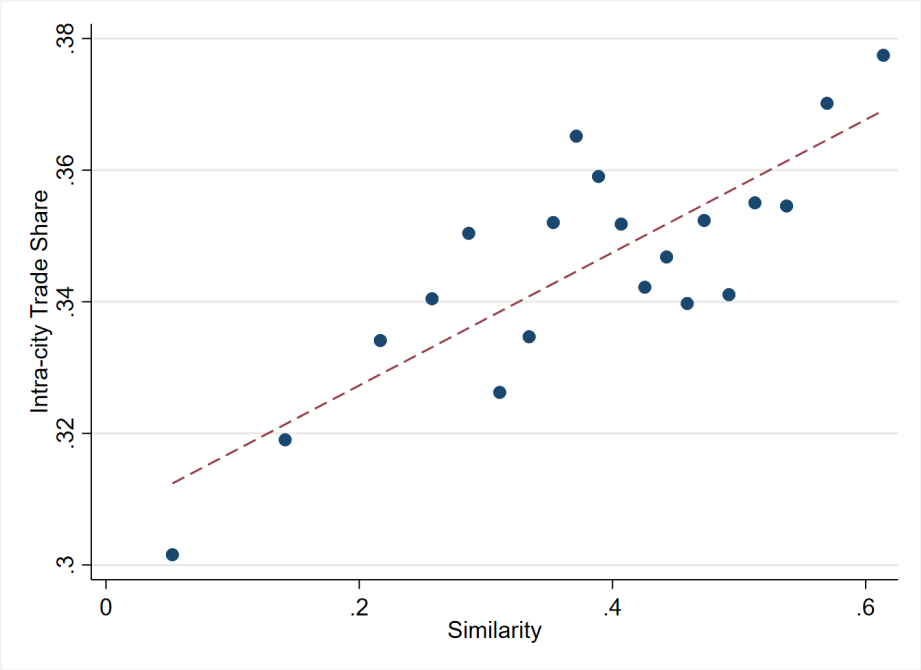
(A) With Provincial Government



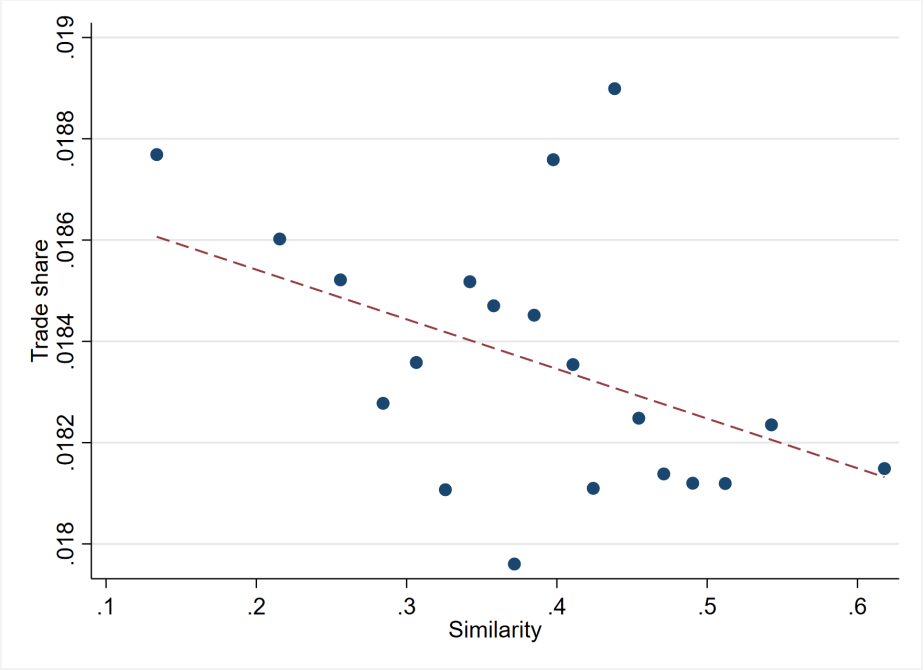
(B) With Central Government

Notes: This figure shows the binscatter plot relating the city's order of entering each sector and the index of tool similarity with provincial government. The unit of observation is city-industry-year. Tool similarity is calculated as the cosine similarity of the tool vector within each city-industry-year cell between city-level government and provincial government.

Figure 9: Policy Sector Similarity and Local Protectionism



(A) City Level



(B) City-pair Level

Notes: Panel (A) is the city-year level binscatter plot relating each city's average policy similarity index and its intra-city trade share. Panel (B) is the city-pair-year level binscatter plot relating each city pair's similarity in policy vector and the trade share between the pair.

Table 1: Comparing LLM vs. Human Label

Panel A: Full Sample			
	RAs Positive	RAs Negative	
LLMs Positive	287	69	Precision: 80.6%
LLMs Negative	79	1565	
	Recall: 78.4%		Accuracy: 92.6%
Panel B: All RAs Agree Subsample			
	RA Positive	RA Negative	
LLMs Positive	186	28	Precision: 87.2%
LLMs Negative	19	1382	
	Recall: 90.7%		Accuracy: 97.1%

Table 2: Industrial Policy Shares, by Government Level

	Industrial Policy			Overall
	#	% in documents	% in all IP	
Central	84,369	19.58	14.16	430,967
Provincial	235,188	18.83	39.46	1,249,034
City	253,757	17.27	42.58	1,469,612
District/County	22,578	17.11	3.79	131,970
Township	65	10.35	0.01	628
Total	595,957	18.16	100.00	3,282,211

Table 3: Industrial Policy Implementation Tools, by Government Levels

	All	Central	Provincial	City
Fiscal and Financial Tools				
Credit and finance	16.85	10.43	14.48	20.65
Tax incentives	16.25	15.83	13.59	18.81
Equity support	3.89	2.42	3.41	4.83
Fiscal subsidies	48.89	25.99	47.15	56.44
Entry and Regulation Tools				
Industrial fund	7.05	3.51	6.14	9.01
Promote entrepreneurship	6.97	2.93	5.69	9.30
Market access and regulation	42.91	53.19	41.31	41.51
Investment policy	13.41	8.44	11.70	16.26
Improving business environment	21.52	14.00	18.43	26.48
Trade protection	5.84	16.93	4.38	3.71
Input Policy Tools				
Labor policy	28.08	17.37	25.76	33.08
Preferential land supply	14.56	5.26	11.12	19.68
Infrastructure	23.89	12.77	21.09	29.07
R&D and tech. adoption	26.70	19.94	26.17	29.40
Environmental policy	19.36	11.42	17.56	22.87
Demand-Side Tools				
Industrial promotion	17.08	9.80	16.75	19.51
Government procurement	6.35	3.77	6.15	7.40
Consumer subsidy	6.02	3.76	5.12	7.45
Supply-Chain Tools				
Industrial cluster	19.07	8.81	16.90	23.84
Localization	3.57	1.70	2.89	4.55

Table 4: Sector Choice and Regional Economic Factors

	(1)	(2)	(3)	(4)
$\text{RCA}^{p}_{cs(t-1)}$	0.0260***	0.0175***	0.0172***	0.0053***
	(0.0003)	(0.0003)	(0.0003)	(0.0008)
$\text{AA}_{cs(t-1)}$	1.3430***	1.1980***	1.1370***	1.1290***
	(0.1430)	(0.1270)	(0.1360)	(0.2540)
$\log(\text{Centrality}^{c}_{cs})$		0.0197***	0.0199***	
		(0.0010)	(0.0010)	
$\log(\text{Centrality}^{p}_{p(c)s})$		0.0083***	0.0084***	
		(0.0010)	(0.0009)	
$\text{Policy}^{Down,+}_{cst}$		0.8000***	0.6480***	0.8130***
		(0.0164)	(0.0157)	(0.0167)
$\text{Policy}^{Up,+}_{cst}$		2.1580***	2.2520***	1.9970***
		(0.0200)	(0.0221)	(0.0205)
City, Industry, Year FE	Yes	Yes	No	No
City-by-Year, Industry-by-Year FE	No	No	Yes	No
City-by-Industry, Year FE	No	No	No	Yes
Observations	2,657,599	2,657,599	1,856,762	1,379,307

Note: This table reports the PPML estimation results of Equation (11). ***, **, and * represent significance at 1%, 5%, and 10%, respectively. The unit of observation is city-industry-year level, and the sample coverage is 2001-2020. The dependent variable is a dummy variable indicating whether each city implements an industrial policy targeting each industry in each year.

Table 5: Vertical Policy Pass-through to Local Governments

	(1)	(2)	(3)	(4)
$\text{Policy}^{p+}_{p(c)st}$	0.4180***	0.2880***	0.1840***	0.2200***
	(0.0063)	(0.0081)	(0.0552)	(0.0557)
Policy^{n+}_{st}	0.2280***	0.2890***	0.2350	0.2970*
	(0.0159)	(0.0227)	(0.1680)	(0.1680)
$\text{Policy}^{p+}_{p(c)st}$*$\text{GDPpc}_{ct}$		-0.0113***	-0.0115***	-0.0112***
		(0.0008)	(0.0008)	(0.0008)
Policy^{n+}_{st}*GDPpc_{ct}		-0.0114***	-0.0116***	-0.0111***
		(0.0021)	(0.0022)	(0.0022)
$\text{Policy}^{p+}_{p(c)st}$*log(# $\text{Cities}_{p(c)}$)			0.0397*	0.0328
			(0.0210)	(0.0211)
Policy^{n+}_{st}*log(# $\text{Cities}_{p(c)}$)			0.0211	0.0152
			(0.0646)	(0.0644)
$\text{Policy}^{p+}_{p(c)st}$*$\text{Connection}_{ct}$				-0.0544***
				(0.0112)
Policy^{n+}_{st}*Connection_{ct}				-0.1210***
				(0.0339)
GDPpc_{ct}		-0.0202***	-0.0199***	-0.0201***
		(0.00240)	(0.00245)	(0.00247)
Connection_{ct}				0.192***
				(0.0337)
City-by-Industry, Y FE	Yes	Yes	Yes	Yes
Observations	1,381,296	1,102,206	1,102,206	1,102,206

Note: This table reports the results of estimating Equation (12). Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 6: Determinants of Political vs. RCA Alignments

	Policy Similarity		RCA Similarity	
	(1)	(2)	(3)	(4)
$\log(\text{LandRevenue}_{ct})$	-0.0003		-0.0006	
	(0.0019)		(0.0016)	
$\log(\text{LandRevenue}_{ct})$*$\mathbb{1}(t \geq 2013)$	-0.0042**		0.0025*	
	(0.0019)		(0.0014)	
$\text{TransferRatio}_{ct}$		0.0050		0.0047
		(0.0197)		(0.0171)
$\text{TransferRatio}_{ct}$*$\mathbb{1}(t \geq 2013)$		0.0408*		-0.0384**
		(0.0228)		(0.0185)
GDPpc_{ct}	-0.0016*	-0.0016	0.0010	0.0014
	(0.0009)	(0.0010)	(0.0009)	(0.0009)
GDPpc_{ct}*$\mathbb{1}(t \geq 2013)$	-0.0003	-0.0012*	-0.0005	-0.0012*
	(0.0005)	(0.0007)	(0.0006)	(0.0006)
Change_{ct}	0.0006	0.0001	0.0008	0.0007
	(0.0026)	(0.0026)	(0.0026)	(0.0026)
Change_{ct}*$\mathbb{1}(t \geq 2013)$	0.0026	0.0052	0.0027	0.0042
	(0.0034)	(0.0038)	(0.0032)	(0.0036)
Connection_{ct}	0.0023	0.0031	0.0027	0.0026
	(0.0042)	(0.0041)	(0.0040)	(0.0044)
Connection_{ct}*$\mathbb{1}(t \geq 2013)$	-0.0115**	-0.0154**	-0.0006	0.0003
	(0.0055)	(0.0063)	(0.0052)	(0.0062)
# Policy_{ct}	0.0037***	0.0037***	0.0018***	0.0018***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
City,Year FE	Yes	Yes	Yes	Yes
Observations	3,676	2,996	3,676	2,996
R-squared	0.9160	0.9190	0.8620	0.8590

Note: This table reports the results of estimating Equation (16). Unit of observation is city-year. Standard errors are clustered at the city level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 7: 2013 as a Turning Point: Political Promotions

	Mayor	Secretary
	(1)	(2)
$\text{Similarity}^{p}_{c(\ell)t}$	0.0551	-0.0998
	(0.116)	(0.116)
$\text{Similarity}^{p}_{c(\ell)t} * \mathbb{1}(t \geq 2013)$	0.185	0.336**
	(0.168)	(0.167)
$\text{Similarity}^{\text{RCA}}_{c(\ell)t}$	0.104	0.214*
	(0.127)	(0.126)
$\text{Similarity}^{\text{RCA}}_{c(\ell)t} * \mathbb{1}(t \geq 2013)$	-0.0537	-0.261
	(0.163)	(0.162)
$\text{Connection}_{\ell t}$	0.105***	0.0295
	(0.0272)	(0.0258)
$\text{Connection}_{\ell t} * \mathbb{1}(t \geq 2013)$	-0.0657	-0.0288
	(0.0402)	(0.0385)
$\text{Age}_{\ell t}$	-0.0119***	-0.0168***
	(0.00221)	(0.00238)
$\text{Tenure}_{\ell t}$	0.0105**	-0.00292
	(0.00498)	(0.00456)
$\log(\text{GDP}_{c(\ell)t})$	0.0957	0.147*
	(0.0760)	(0.0753)
$\log(\text{Population}_{c(\ell)t})$	0.0976	-0.0717
	(0.0983)	(0.0974)
City, Province-by-Year FE	Yes	Yes
Observations	4,264	4,260
R-squared	0.446	0.437

Note: This table reports the results of estimating Equation (18). Unit of observation is city-year. Standard errors are clustered at the city level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 8: Effects of Economic and Political Alignments on Firm Entry and Firm Performance

	log(New Capital)	TFP	log(Profit)	log(Wage)	Product Markup	Wage Markdown
	(1)	(2)	(3)	(4)	(5)	(6)
Policy^{+}_{cst}	0.1230***	0.0109**	0.0627***	0.0037***	-0.0129***	0.0071
	(0.0131)	(0.0045)	(0.0093)	(0.0004)	(0.0023)	(0.0066)
$\text{RCA}^{p}_{cs(t-1)}$	5.13e-06**	0.0029***	0.0035**	0.0045***	0.0014***	0.0027**
	(0.0000)	(0.0008)	(0.0016)	(0.0001)	(0.0004)	(0.0013)
Policy^{+}_{cst}*$\text{RCA}^{p}_{cs(t-1)}$	0.0329***	0.0007	0.0007	0.0002**	0.0015***	-0.0021***
	(0.0016)	(0.0006)	(0.0014)	(0.0001)	(0.0003)	(0.0008)
$\text{Policy}^{p+}_{p(c)st}$	0.1570***	-0.0043**	0.0119***	-0.0067***	0.0027**	-0.0208***
	(0.0039)	(0.0022)	(0.0046)	(0.0002)	(0.0011)	(0.0033)
Policy^{+}_{cst}*$\text{Policy}^{p+}_{p(c)st}$	0.1040***	-0.0101**	-0.0340***	-0.0047***	-0.0115***	0.0132*
	(0.0140)	(0.0046)	(0.0097)	(0.0005)	(0.0024)	(0.0069)
City-by-Industry, Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,896,320	3,251,700	4,239,100	4,594,400	2,354,300	2,285,800

Note: This table reports the results of estimating Equation (19). Unit of observation is city-industry-year for column (1) and firm-year for columns (2)-(6). Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 9: Policy Diffusion and Overcapacity: Firm Entry and Performance

	log(New Capital)	TFP	log(Profit)	log(Wage)	Product Markup	Labor Markdown
	(1)	(2)	(3)	(4)	(5)	(6)
Policy^{+}_{cst}	0.0608***	0.0127***	0.0362***	0.0029***	-0.0019	-0.0160***
	(0.0091)	(0.0024)	(0.0049)	(0.0002)	(0.0013)	(0.0034)
Policy^{+}_{cst}*Order_{cs}	0.0552**	-0.103***	-0.0338*	-0.0037***	-0.0143**	0.104***
	(0.0260)	(0.009)	(0.0176)	(0.0009)	(0.0057)	(0.0141)
City-by-Industry, Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,896,320	3,414,551	4,472,606	4,594,416	2,189,634	2,189,564
R-squared	0.726	0.126	0.318	0.991	0.088	0.242

Note: This table reports the results of estimating Equation (20). Unit of observation is city-industry-year for column (1) and firm-year for columns (2)-(6). Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 10: Policy Persistence

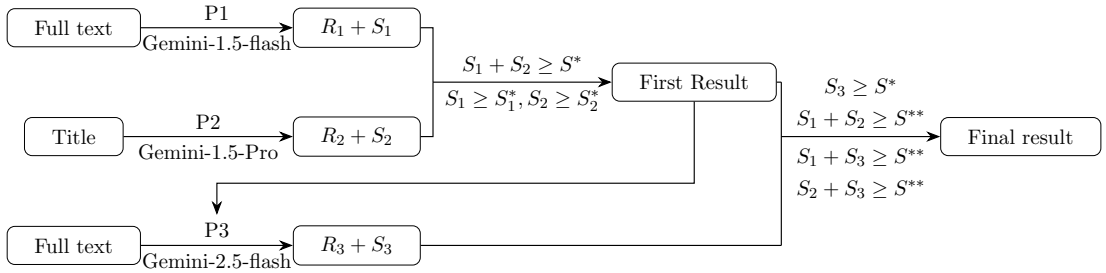
	1 Year		2 Year		3 Year	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_k \log(\text{Capital}_{cst})$	0.0126**	0.0108	0.0118**	0.0293	0.00715*	-0.0271
	(0.00581)	(0.0221)	(0.00467)	(0.0204)	(0.00422)	(0.0235)
$\text{Policy}^{p+}_{p(c)s(t-k)}$	0.0600***	0.124***	0.0653***	0.136***	-0.00189	-0.0148
	(0.0131)	(0.0275)	(0.0134)	(0.0310)	(0.0133)	(0.0345)
$\text{Policy}^{p+}_{p(c)st}$	0.491***	0.550***	0.513***	0.563***	0.539***	0.618***
	(0.0140)	(0.0305)	(0.0157)	(0.0383)	(0.0170)	(0.0489)
$\Delta_k \log(\text{Markdown}_{cst})$		-0.00487		0.0168**		0.0190**
		(0.00643)		(0.00718)		(0.00880)
$\Delta_k \log(\text{Markup}_{cst})$		-0.0165		-0.0347**		-0.0110
		(0.0130)		(0.0155)		(0.0193)
City-by-Industry, Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	77,507	13,239	66,584	6,490	57,544	9,524

Note: This table reports the results of estimating Equation (21). Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

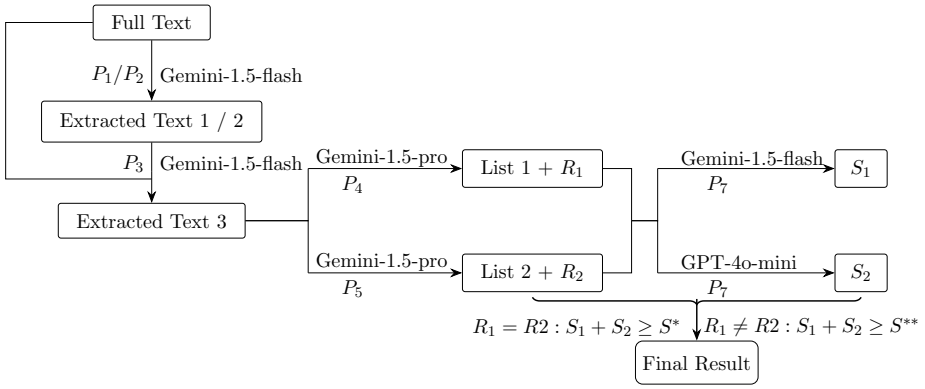
Online Appendix (Not For Publication)

Appendix A Additional Figures and Tables

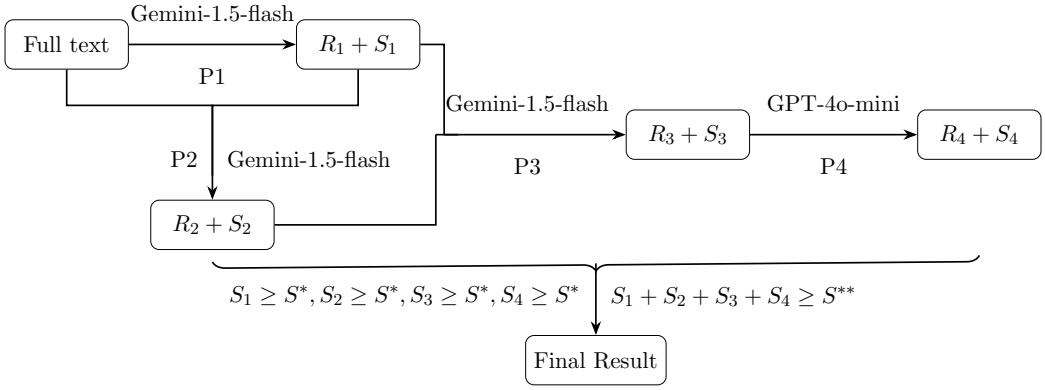
Figure A1: LLM Implementation Roadmap



(A) Determine Industrial Policy



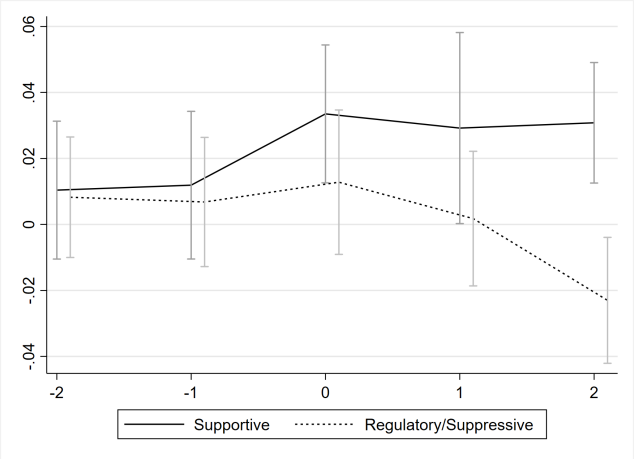
(B) Policy-targeted Industry



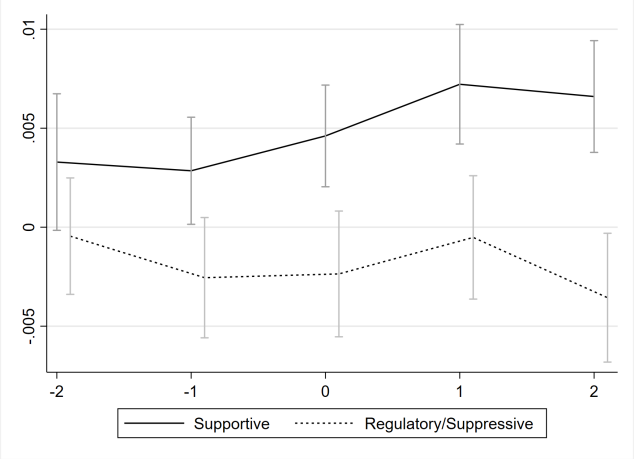
(C) Policy Tool and Implementation

Notes: This figure presents the roadmap for LLM implementation. Pn represents the prompt for the n-th round of LLM inquiry. R_n represents the LLM response from the n-th round, including the extracted relevant text; S_n represents the confidence score of the n-th round LLM response. S^* represents the threshold score for final determination, which can be adjusted flexibly.

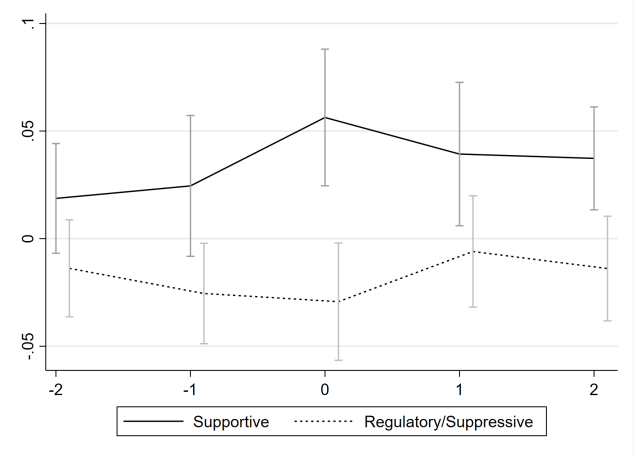
Figure A2: Effect of Industrial Policy: Subsidy, Tax Deduction, and Long Term Debt



(A) Subsidy



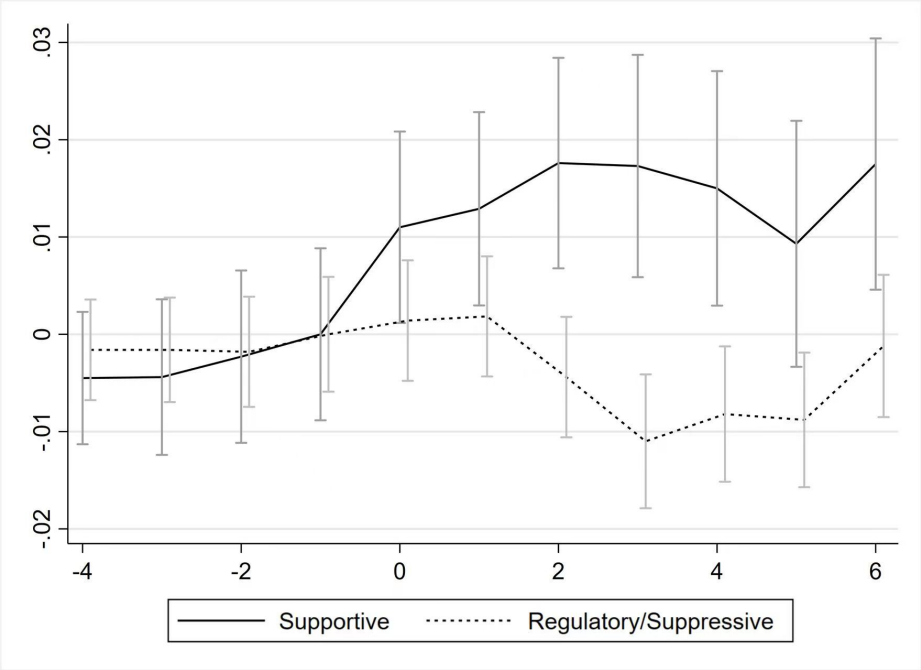
(B) Tax Deduction



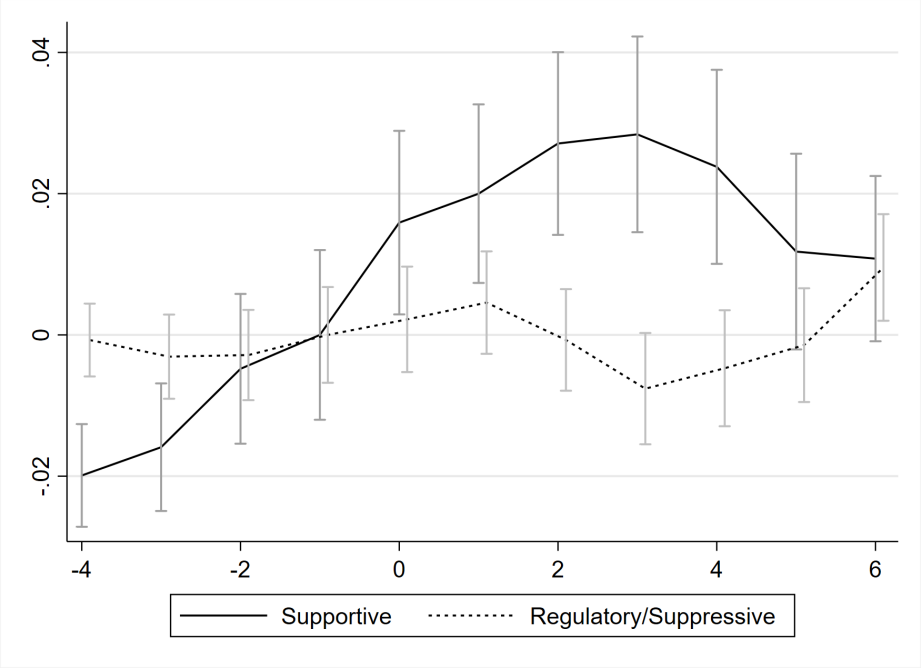
(C) Long Term Debt

Notes: This figure plots the coefficients and the corresponding 95% confidence intervals of estimating Equation (1). Unit of observation is firm-year. Standard errors are clustered at the city-by-industry level.

Figure A3: Dynamic Effects of Supportive vs. Regulatory Policies on New Firm Entry



(A) City × Industry, Year Fixed Effects



(B) City × Industry × 5-Year-Plan, Year Fixed Effects

Notes: This figure plots the coefficients and the corresponding 95% confidence intervals of estimating Equation (2). Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level.

Figure A4: Spatial Distribution of the Industrial Policies

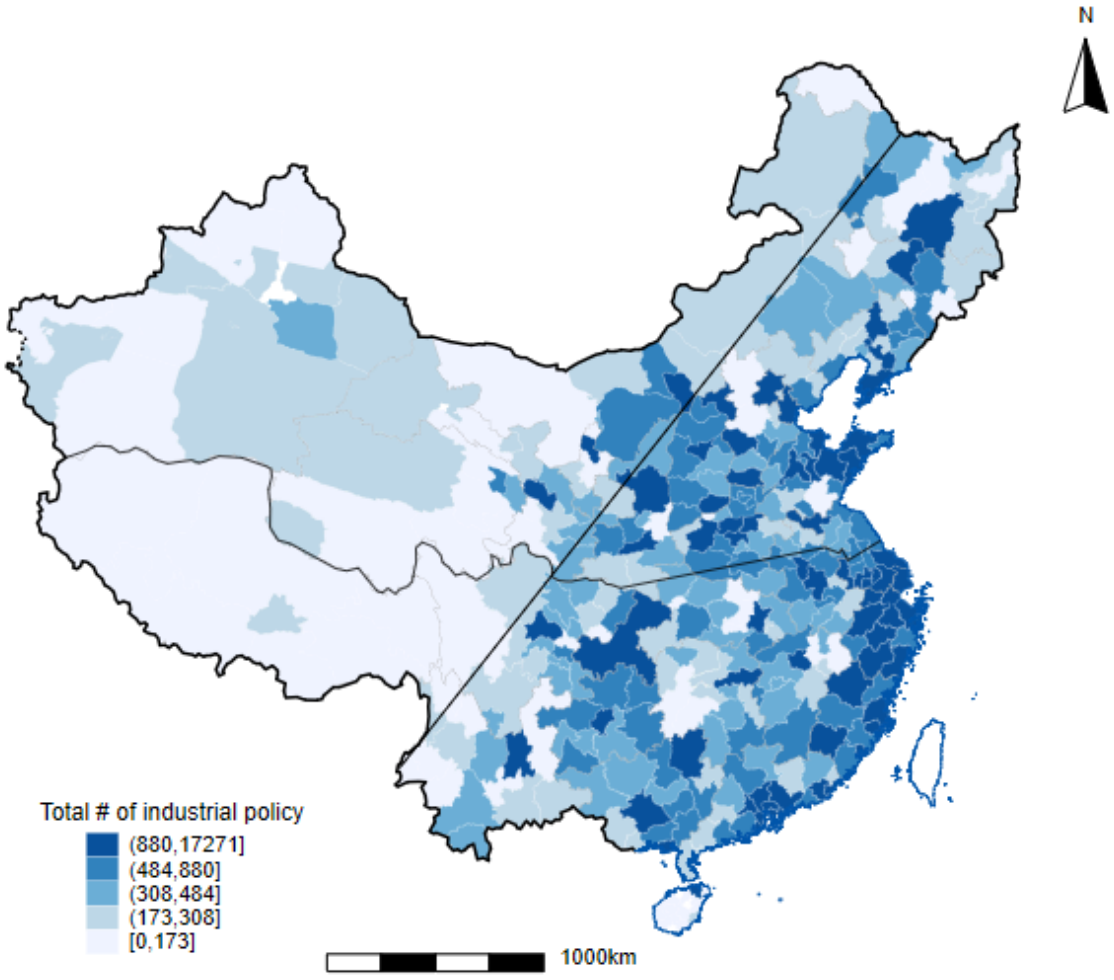
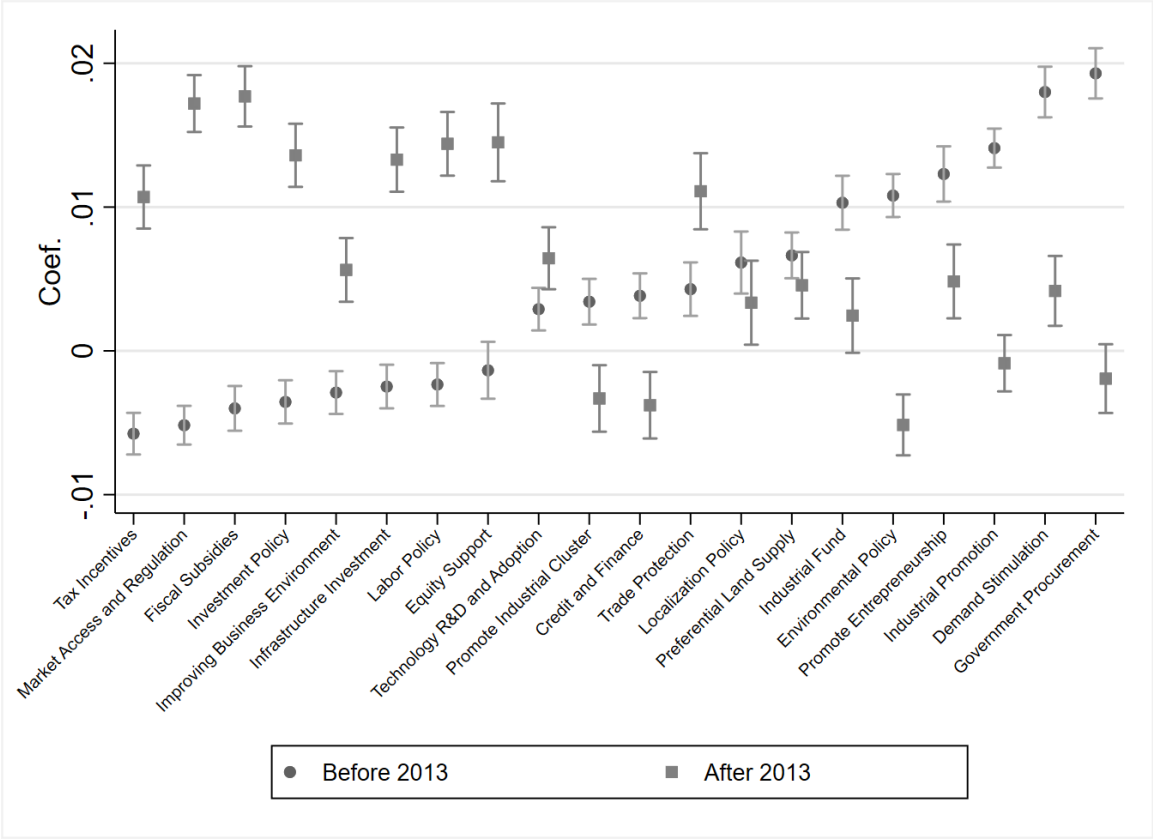


Figure A5: Effect of Provincial Policy Tools on Top-Down Alignment: Before and After 2013



Notes: This figure plots the coefficients and the corresponding 95% confidence intervals for each industrial policy tool by estimating Equation (17). Unit of observation is city-industry-year level. Standard errors are clustered at the city-by-industry level.

Table A1: Comparing LLM versus BERT versus Human Label

Method	Precision	Recall	F1-score	Accuracy
Logistic-L1	0.4898	0.7394	0.5841	0.8033
Logistic-L2	0.7466	0.5273	0.6039	0.8744
Logistic-Elastic Net	0.5128	0.7273	0.5983	0.8211
Random Forest	0.5765	0.5818	0.5684	0.8378
BERT (Layer-optimized)	0.5423	0.5515	0.5384	0.8244
BERT (Fully-optimized)	0.6432	0.703	0.6708	0.8733
LLM	0.8066	0.7576	0.7813	0.9222

Table A2: Industrial Policy Objectives, by Government Level

	All	Central	Provincial	City
Key Industry				
Promote strategic industry	26.67	21.42	26.19	29.13
Promote pillar industry	5.64	2.90	5.27	6.57
Promote emerging industry	13.05	7.28	11.41	16.39
Support traditional advantageous industry	12.72	9.98	13.89	12.07
Upgrade traditional industry	15.67	9.74	15.14	17.72
Support green industry	15.91	10.40	15.58	17.60
Promote other key industry	11.20	9.37	11.33	11.50
Innovation				
Promote innovation	19.72	15.44	20.05	21.21
Promote new technology adoption	10.63	6.83	9.69	12.64
Social Welfare				
Promote social equity and welfare	28.58	19.46	27.02	31.59
Urbanization	5.53	1.92	3.50	8.38
Stimulate employment	13.41	8.62	12.15	15.60

Table A3: Industrial Policy Conditionality, by Government Level

	All	Central	Provincial	City
Firm location	44.93	30.99	45.51	48.47
Specific firms	14.96	15.14	15.22	14.60
R&D investment or specific technology	14.29	10.44	13.99	16.10
Firm age	9.61	6.87	8.88	11.13
Firm ownership type	13.64	15.44	12.29	14.28
Firm scale	38.25	25.30	35.56	44.36

Table A4: Summary Statistics for Firm Registration

Variable	Mean	Std. dev.	Min	Max
Panel A: Firm Registration				
New registration capital (billion RMB)	191.28	14,226.42	0	16,057.89
New registration # of firms	26.83	381.76	0	118,629.00
Cumulative registration capital (billion RMB)	1,425.19	40,109.63	0	16,238.90
Cumulative registration # of firms	229.16	2,330.35	0	535,472.00
RCA^p	1.29	9.70	0	455.78
RCA^n	1.55	14.50	0	3,975.96
AA	0.00	0.01	0	1.00
Panel B: Supply Chain				
$\log(\text{Centrality}^c)$	10.48	8.67	0	29.22
$\log(\text{Centrality}^p)$	10.46	8.82	0	29.45
$\text{Policy}^{Up,+}$	0.13	0.19	0	1
$\text{Policy}^{Down,+}$	0.12	0.17	0	1

Table A5: Summary Statistics for Administrative Tax Data

	Obs.	Mean	SD	Min	Max
Subsidy (thousand RMB)	3,760,219	349.61	1,936.95	0	20,727
Tax deduction rate	6,080,762	0.07	0.23	0	1
Leverage (equity/asset)	7,027,631	0.66	0.64	-0.07	6.27
1(Long term debt)	7,239,822	0.07	0.26	0	1
Revenue (thousand RMB)	7,012,096	121,218.30	447,571	1	4,508,752
Output value (thousand RMB)	5,361,358	84,565.97	323,932.10	0	3,262,687
Value-added (thousand RMB)	4,499,134	15,736.92	67,731.90	-11257	688,829
Input value (thousand RMB)	4,391,739	70,188.01	273,349.80	0	2,766,013
Employment	7,170,999	111.66	302.69	1	2797
Log(TFP) (revenue-based)	5,957,269	6.07	1.73	-19.92	18.02
Wage (thousand RMB)	6,684,833	5254.53	14993.41	2	109,344
Product Markup	2,220,049	1.07	0.58	0.21	4.35
Labor Markdown	2,219,977	11.16	24.99	.48	180.54

Table A6: Summary Statistics for City Politicians

	Secretary			Mayor		
Variable	Obs	Mean	Std. Dev.	Obs	Mean	Std. Dev.
Promotion	5,650	0.11	0.32	5,652	0.20	0.40
Age	5,628	52.62	3.63	5,641	50.63	3.87
Male	5,642	0.96	0.18	5,647	0.94	0.24
Tenure	5,660	2.72	1.64	5,660	2.58	1.56
Change	5,660	0.28	0.45	5,660	0.29	0.46
Connection with prov. secretary	5,660	0.09	0.29	5,660	0.06	0.24
Connection with governor	5,660	0.11	0.32	5,660	0.11	0.32

Table A7: Sector Choice and Regional Advantage

	(1)	(2)	(3)
$\text{RCA}^{p}_{cs(t-1)}$	0.00861***	0.00628***	0.00956***
	(0.000443)	(0.000765)	(0.000516)
$\text{AA}_{cs(t-1)}$	3.636***	2.789***	2.721***
	(0.187)	(0.210)	(0.173)
$\log(\text{Centrality}^{c}_{cs})$		0.0167***	0.0162***
		(0.00183)	(0.00175)
$\log(\text{Centrality}^{p}_{p(c)s})$		0.0106***	0.0102***
		(0.00177)	(0.00170)
$\text{Policy}^{Down,+}_{cst}$		1.422***	1.260***
		(0.0368)	(0.0348)
$\text{Policy}^{Up,+}_{cst}$		3.003***	3.010***
		(0.0441)	(0.0484)
City, Industry, Year FE	Yes	Yes	No
City-by-Year, Industry-by-Year FE	No	No	Yes

Note: This table reports the PPML estimation results of Equation (11) excluding the sample of subsequent years after each industry first being supported in each city. Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table A8: Interaction between Economic and Political Factors in Sector Choice

	(1)	(2)	(3)
$\text{Similarity}^{p}_{ct}$	3.541***		3.574***
	(0.0368)		(0.0534)
$\text{RCA}^{p}_{cs(t-1)}$	0.0132***	0.0177***	0.00615***
	(0.00142)	(0.00159)	(0.00196)
$\text{RCA}^{p}_{cs(t-1)}$*$\text{Similarity}^{p}_{ct}$	-0.00808**	-0.00482	-0.00908*
	(0.00352)	(0.00397)	(0.00467)
$\text{AA}_{cs(t-1)}$	4.025***	4.288***	3.799***
	(0.184)	(0.205)	(0.375)
$\text{AA}_{cs(t-1)}$*$\text{Similarity}^{p}_{ct}$	-3.899***	-4.734***	-5.788***
	(0.395)	(0.449)	(0.697)
$\log(\text{Centrality}^{c}_{cs})$	0.0551***	0.0532***	
	(0.00271)	(0.00291)	
$\log(\text{Centrality}^{c}_{cs})$*$\text{Similarity}^{p}_{ct}$	-0.0794***	-0.0750***	-0.0588***
	(0.00522)	(0.00573)	(0.00926)
$\log(\text{Centrality}^{p}_{p(c)s})$	-0.00316	-0.00474*	
	(0.00259)	(0.00278)	
$\log(\text{Centrality}^{p}_{p(c)s})$*$\text{Similarity}^{p}_{ct}$	0.0311***	0.0336***	0.00778
	(0.00499)	(0.00547)	(0.00883)
$\text{Policy}^{Down,+}_{cst}$	1.739***	1.920***	1.496***
	(0.0421)	(0.0436)	(0.0458)
$\text{Policy}^{Down,+}_{cst}$*$\text{Similarity}^{p}_{ct}$	-2.384***	-2.577***	-1.989***
	(0.0693)	(0.0741)	(0.0756)
$\text{Policy}^{Up,+}_{cst}$	3.721***	4.596***	3.565***
	(0.0508)	(0.0582)	(0.0555)
$\text{Policy}^{Up,+}_{cst}$*$\text{Similarity}^{p}_{ct}$	-3.672***	-4.440***	-4.057***
	(0.0803)	(0.0969)	(0.0874)
$\text{Policy}^{p+}_{p(c)st}$	-0.145***	-0.213***	-0.642***
	(0.0127)	(0.0140)	(0.0132)
$\text{Policy}^{p+}_{p(c)st}$*$\text{Similarity}^{p}_{ct}$	1.344***	1.331***	2.114***
	(0.0237)	(0.0265)	(0.0240)
City,Industry,Year FE	Yes	No	No
City-by-Year, Industry-by-Year FE	No	Yes	No
City-by-Industry, Year FE	No	No	Yes
Observations	1,938,566	1,858,688	1,085,889

Note: This table reports the results of estimating Equation (11) augmented with $Similarity^{p}_{ct}$ interactions. Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table A9: Policy Diffusion and Sector Choice

	(1)	(2)	(3)
$RCA^{p}_{cs(t-1)}$	0.0176***	0.0226***	0.00651***
	(0.000744)	(0.000795)	(0.00167)
$RCA^{p}_{cs(t-1)}$*$Order_{cs}$	-0.0243***	-0.0231***	-0.0221***
	(0.00279)	(0.00242)	(0.00432)
$Policy^{p+}_{p(c)st}$	-0.136***	-0.127***	0.0891***
	(0.00881)	(0.00873)	(0.00907)
$Policy^{p+}_{p(c)st}$*$Order_{cs}$	3.035***	2.550***	1.595***
	(0.0347)	(0.0336)	(0.0338)
$Order_{cs}$	-4.828***	-4.371***	
	(0.0352)	(0.0339)	
City, Industry, Year FE	Yes	No	No
City-by-Year, Industry-by-Year FE	No	Yes	No
City-by-Industry, Year FE	No	No	Yes
Observations	2,661,240	1,859,526	1,381,660

Note: This table reports the results of estimating Equation (11) with additional interaction terms with $Order_{cs}$. Unit of observation is city-industry-year. Standard errors are clustered at the city-by-industry level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Appendix B Details of LLM Implementation: Overcoming Hallucination

B.1 Large Language Models.

LLMs are deep learning models with billions of parameters, which makes them well-suited to text tasks such as summarization, extraction, and classification. They acquire linguistic and factual knowledge from large, diverse text corpora and prediction tasks, and are fine-tuned to follow human instructions through feedback. Despite these strengths, LLMs can generate incorrect responses, termed "hallucination," when they are uncertain about an answer or misinterpret instructions. An LLM can be understood as a broadly-read research assistant: capable across many domains, but prone to confident errors. Clear instructions and strategies to minimize hallucinations are therefore essential for accurate outcomes.

We mainly utilize Google's `Gemini-1.5-flash-001` model from DeepMind. We choose the model based on two major reasons. First, our text analysis tasks require extensive knowledge and proficiency in Chinese. The model is trained on a large multilingual corpus totaling 6.5 trillion tokens, equivalent to millions of books, providing a solid knowledge foundation for our analyses. User feedback indicates that `Gemini-1.5-flash-001` outperforms comparable versions of `GPT-4`, such as `GPT-4-0314` and `GPT-4-0613`, and most Chinese LLMs in text analysis tasks conducted in both English and Chinese. Second, the model can process texts of up to one million tokens (roughly 750,000 words) without noticeable degradation in performance, which allows us to supply both complete policy texts and the relevant definitions and guidance in a single query. This capacity is important because many of our queries require comprehension of the full document, and their resolution often depends on clear definitions and guidance. For example, determining if a policy document qualifies as industrial policy requires assessing several factors: government issuance, industry targeting, the inclusion of policy tools, and intent to alter the long-term economic structure. Because the relevant information is dispersed across the entire policy document, full-text querying of the LLMs is critical. Moreover, discerning whether a policy targets specific industries or not, and whether it aims to change the economy's structure versus responding to short-term shocks, both involve nuanced judgment. It is thus important to provide clarifying definitions in more detail and step-by-step guidance with a list of counterexamples, which are shown to significantly enhance output quality. Nevertheless, our policy documents are lengthy, with an average of 1,160 words and a maximum of 5,461 words. Also, defining industrial policy and tools alone takes 2,406 words. Consequently, the `Gemini-1.5-flash-001` model fits our needs.

B.2 Major Prompt Strategies.

To reduce hallucinations in LLM outputs, we utilize various prompt design methods to boost accuracy. First, we provide detailed and clear definitions to minimize ambiguity. We include counterexamples where applicable to clarify understanding. The counterexamples are summarized through iterative prompt experiments using different policy document samples. Second, we present structured guidelines for text analysis and break down complex tasks into step-by-step questions to enhance LLMs' analytical abilities. For instance, before evaluating whether a policy document is an industrial policy, the LLM must answer eleven detailed questions related to industrial policy, including the policy issuing identity, the policy's sector target, policy objective, and related reasoning. Third, we employ general prompt strategies from the literature: setting the temperature to zero for consistency,[21] assigning the LLM a role as a China industrial policy expert, emphasizing answering based on the text, allowing "I don't know" responses, emphasizing conservative judgment to reduce false classifications, ensuring consistency among questions, using clear formatting, and stressing key points. Importantly, we require answers to major questions to include confidence scores (0-100), concise reasoning, and all relevant text excerpts. These requirements encourage the LLMs to maintain logical consistency and mitigate hallucinations. In addition, we can directly mitigate hallucinations for relevant tasks by verifying whether the policy texts contain extracted excerpts. Moreover, as described below for policy targets and policy tools, this approach makes it convenient for another LLM to critique initial responses for further improvement. Lastly, providing relevant text and reasoning further improves transparency.

B.3 Overview of the Processes.

We conduct two major rounds of LLM queries using both the titles and the full texts of policy documents, with each round containing several rounds of refinement. The first round covers all policy documents in our sample. The second round focuses on a subset tentatively identified as industrial policies, which we detail in the next section. We separated the queries into two rounds because the performance of the LLM deteriorates with excessively long questions, despite the strong performance of the model in processing long texts.

In the first major round, our queries address several key aspects: whether the policy document constitutes an industrial policy; the issuing government; the list of targeted industries; policy tone; policy objectives; the list and classification of policy tools; measures of policy strength; the conditionality of the policy; references to other governments; and citations of other policies. The second major round of queries delves into additional dimensions and repeats some first-round questions to enhance the output quality: policy issuance dates and effective periods; the list of targeted industries; policy tone; organizational arrange-

[21] In artificial intelligence (AI) and machine learning, temperature is a parameter for adjusting the output of LLMs. Temperature controls the randomness of text generated by LLMs during inference. Except for temperature, the other parameters are set to default.

ments for policy implementation, including the role of political incentives, experimentation and learning, adaptation to local conditions, and organizational support in policy implementation; and continued references to other governments and citations of other policies. Appendix C provides a detailed list of the main components of the prompts we query the LLM.

B.4 Multi-round Refinement with Additional LLM Analysis.

Building on the LLM agent framework (Madaan et al., 2023; Shinn et al., 2023; Li et al., 2024), we further mitigate hallucination by integrating independent judgments from multiple LLMs. Specifically, after the first round of LLM output, we employ an additional LLM to critically review the outputs, summarize common errors in the initial round of LLM outputs, and construct prompts enriched with examples and counterexamples for subsequent rounds. We involve human experts in the loop to review the critiques and ensure the accuracy of the summary, examples, and counterexamples that we provide to subsequent rounds of LLMs. We iterate the process when there are noticeable discrepancies between the first two rounds until convergence. In each round, we require the LLM to respond with references from the original document, reasoning, and a confidence score. The reasoning and confidence scores are used to refine LLM outputs and determine the final classification. We apply this approach to several critical tasks, including the determination of industrial policy, the mapping of texts to industry codes, and the classification of policy implementation tools.

Determination of Industrial Policy. We conduct three rounds of LLM to determine whether a policy document is an industrial policy, the procedure of which is summarized in Panel (A) of Figure A1. As a first step, we provide LLM with a clear definition of industrial policy along with step-by-step questions to guide the LLM through each criterion. To summarize, a document qualifies as industrial policy if it meets four criteria:

1. The policy issuer must be a *government entity* at any level; texts involving only firms or non-governmental organizations are excluded.
2. The policy must contain *concrete government actions* rather than general statements or progress reports.
3. The policy must be explicitly *biased toward* (for or against) specific industries; broad macroeconomic programs without targeted sectors are excluded.
4. The policy must aim at *long-term* structural transformation rather than short-term stabilization, such as responses to COVID-19 or cyclical downturns.

After the initial round, we conduct two additional rounds of LLM evaluations. For the second round, we use a more advanced yet expensive LLM, `Gemini-1.5-pro-exp-0827`, developed by Google DeepMind, which surpasses `Gemini-1.5-flash-001` in knowledge

and ability to handle complex tasks. We first extract a random sample of 10,000 policy documents that are determined to be industrial policies from the first round LLM. We ask the more advanced model to critically review the documents and summarize the potential common mistakes. We manually review the summary to construct a list of counterexamples that frequently cause confusion, and then iterate the process until we do not find any more types of mistakes. This two-step method ensures a comprehensive and accurate summary of all common non-industrial-policy documents, thereby minimizing misclassifications. We take a cautious stance, focusing only on obvious counterexamples to mitigate the chances of misclassifications. In this round of the prompt, we reiterate the definition of industrial policies and provide the LLM with this manually checked list of counterexamples, instructing the LLM to utilize all relevant information to arrive at the final decision. To manage costs, we only provide it with the policy titles, which are the most indicative of the policies' nature. In both rounds of LLM queries, we instruct the LLM to respond with reasoning and confidence score, denoted S_1 and S_2, respectively.

For the third round, we employ `Gemini-2.5-flash`, a model that offers performance comparable to `Gemini-1.5-pro` at a much lower cost, allowing, once again, for a full-text analysis.[22] We apply it to all full policy texts. The prompt includes examples and counterexamples of industrial policies, which we obtain by summarizing and manually reviewing the outputs of the first two rounds. The third round also outputs a confidence score, denoted S_3.

Finally, we combine the first round's LLM scores with the subsequent two rounds' to classify whether a document constitutes an industrial policy. Specifically, our criteria are mainly: 1) the third round LLM score is at least 90, i.e. $S_3 \geq 90$ or 2) the sum of any two rounds of scores is at least 180, i.e. $S_1 + S_2 \geq 180$ or $S_1 + S_3 \geq 180$ or $S_2 + S_3 \geq 180$. This criterion effectively places a higher weight on the third round LLM score because it utilizes a powerful model, `Gemini-2.5-flash` , on the policy full text.[23] Through these procedures, we identify 595,957 industrial policies out of a total of 3.28 million policy documents.

Extracting Industries and Matching to Standardized Industry Codes. To accurately identify the targeted industries from policy texts, we follow a multi-stage procedure to improve the performance of LLMs, which is summarized in Panel (B) of Figure A1. As explained in Section B.3, we conducted two main rounds of LLMs queries, and in both rounds, we prompted the LLMs to extract industry-related text using full policy documents. We avoid direct mapping of full policy texts to complete industry lists due to potential performance issues with long texts. The full list of industry names with policy texts involves more than 1,000 lines of industry code and industry names; thus, mapping the full text with

[22]The model was not released at the time we ran the first round and only became available as our project proceeded to the third round.

[23]In principle, we can adjust the thresholds in the three rounds. The current thresholds are based on our experiments using random samples of documents through several iterations. They are relatively stringent.

the full list would notably increase the input length and hallucination risks. Moreover, the extracted text allows for further refinement. The two rounds of text extraction are largely the same, which assures us of the accuracy of the responses.

One potential issue we identify from human review of a random sample is that policy texts typically mention multiple industries–some as main targets and others briefly; mapping all of them to industry codes as the targeted industries of the policy will lead to false positives. To distinguish between these, we employ an additional LLM to review the texts extracted from both rounds alongside the complete policy texts, and ask it to add any missed relevant information, remove errors due to hallucinations, and identify main policy targets from briefly mentioned industries. We then provide the refined list of relevant texts regarding main policy targets and a full list of standardized industry codes and industry names to the advanced LLM, `Gemini-1.5-pro-exp-0827`. This model, acting as an expert on Chinese industries, maps extracted texts to standardized industry codes.[24] We matched each text description to industry codes at two levels of granularity, using both three- and four-digit code lists. This hierarchical procedure serves two purposes. First, the two rounds of mapping provide an internal consistency check, as the four-digit code should fall within the corresponding three-digit category. Second, it reduces false precision: when a description refers only to a broad industry, a three-digit classification is more defensible than forcing it into an overly specific four-digit code. For each candidate mapping, we ask the LLM to reason whether the industry is a major target of the policy or is only briefly mentioned by the policy document.

Lastly, to further improve matching accuracy, we conduct two rounds of review of each candidate standardized industry code mapping using `Gemini-1.5-flash` and `GPT-4o-mini`, with LLM-rated scores denoted as S_1 and S_2 respectively. We combine results from the two rounds of mapping and two rounds of reviews to generate the final result of the list of targeted industries. Specifically, we keep the match when both mapping models agree that the industry is the policy's major target and the two reviewers do not simultaneously reject it (the sum of reviewer scores ¿=10), or when only the four-digit industry mapping model lists it as a weak industry match and the two reviewer models achieve strong agreement (the sum of review scores ¿=150).

Policy Tools. The extraction and categorization of policy implementation tools also follow a multi-stage procedure. We organize policy tools into 20 categories. To improve precision, we provide LLMs with explicit definitions for each policy tool, as detailed in Section C. As with the classification of the targeted industry, we separate the text extraction task from the classification task to reduce the risk of hallucination. We first ask the LLM to extract policy tool related text and then ask it to map the text into relevant categories.

[24] A small fraction of texts was processed using the standard-release version of the same `Gemini-1.5-pro` model due to changes in the availability of specific model versions. These two versions yielded consistent results in this industry classification task.

The mapping is not limited to one category to prevent false negatives. In particular, we ask the LLM to match the keywords one by one to each category.

A manual review of the first round of LLM outputs shows high extraction accuracy; however, some misclassifications occur. In the second round of LLM query, we further refine the results by providing the LLM with more detailed definitions and counterexamples. We obtain the counterexamples following the same procedures as in the determination of industrial policies—we have an LLM summarize outcomes using a 10,000 random sample from the first round and then manually refine these summaries. We then feed the extracted relevant texts, as well as LLM outputs from the first two rounds, along with the definitions and counterexamples, to the third round of LLM queries for the model to conduct self-critique; and feed the first three rounds of responses to a more orthogonal model, `GPT-4o-mini` , for one more round of refinement. The final classification combines the results from the four rounds.

Figure A1 summarizes our discussions above and illustrates the broad roadmap of our LLM approach to determining industrial policies, mapping to industry codes, and classifying policy tools.

[Figure A1 about here]

Appendix C Major Components of LLM Prompts

This section presents the summary of the definitions we provide to the LLMs for the major classification tasks.

C.1 Policy Objectives

1. **Promote strategic industry**: Policies aimed at fostering sectors deemed critical for national or regional development.
2. **Promote emerging industry**: Supporting nascent sectors like EV or artificial intelligence (AI) that are expected to drive future economic growth.
3. **Promote pillar industry**: Supporting the traditional sectors that form the backbone of the economy, such as steel, machinery, or textiles.
4. **Support traditional advantageous industry**: Supporting traditional industries that are important for the local economy with established advantages.
5. **Upgrade traditional industry**: Encouraging traditional sectors to modernize and adopt new technologies in order to remain competitive in a globalized market.
6. **Support green industry**: Supporting sectors involved in environmental sustainability, renewable energy, and green technologies to foster a low-carbon economy.

7. **Support other key industries**: Supporting sectors of key importance that do not belong to the above listed categories.
8. **Promote innovation**: Policies aimed at developing technologies that address critical technological challenges, such as "choke point" technologies or frontier tech advancements.
9. **Promote new technology adoption**: Encouraging firms to adopt cutting-edge technologies, such as digitization or artificial intelligence, to enhance productivity and competitiveness.
10. **Stimulate employment**: Industrial policies that focus on stimulating labor-intensive industries, mitigating unemployment, and providing a safety net for displaced workers.
11. **Promote social equity and welfare**: Policies that prioritize inclusive development, poverty alleviation, rural revitalization, or improving social welfare.
12. **Urbanization**: Supporting policies that promote urban development, integrate rural and urban areas, and manage the challenges associated with rapid urbanization.

C.2 Industrial Policy Tool

1. **Tax incentives:** Implementing industrial policies through tax reductions, tax credits, accelerated depreciation, etc.
2. **Fiscal subsidies:** The government subsidizes companies or industries through direct fiscal subsidies, in-kind grants, production subsidies, etc., as a form of industrial policy.
3. **Credit and finance:** The government guides financial institutions to provide low-interest loans, interest subsidies, loan guarantees, development bank loans, policy-directed lending by financial institutions, and priority bank credit to industries or enterprises as a form of industrial policy.
4. **Equity support:** State-owned enterprises, governments, local government financing platforms, etc., support the development of specific industries or enterprises through equity injections.
5. **Industrial fund:** The government establishes industrial funds or government investment funds to provide venture capital support, invest through industrial funds, guide government investments, prioritize government venture capital support in specific industries.
6. **Trade protection:** The government implements industrial policies through export bans, export licensing requirements, export quotas, export taxes, export subsidies, export tax incentives, export credit agencies, export-related non-tariff measures, other

export incentives, import bans, import licensing requirements, import monitoring, import quotas, import tariffs, import-related non-tariff measures, internal taxes on imports, technical barriers to imports, technical matching or technology transfer requirements related to imports, anti-dumping measures, countervailing measures, and other import incentives. Similar policies may also be used to target enterprises from other regions rather than foreign enterprises.

7. **Investment policy:** The government attracts and guides foreign investment by imposing entry restrictions on foreign or out-of-region enterprises, shareholding ratio restrictions on foreign or out-of-region enterprises, preferential policies for foreign or out-of-region investments; establishing specialized investment promotion agencies to provide investment consulting, project matchmaking, policy interpretation, and other services.
8. **Technology R&D and adoption:** The government encourages enterprises to conduct R&D through R&D subsidies, funding coordination, technology innovation funds, additional deductions for R&D expenses, support for technology transfer, industry-university-research cooperation alliances, establishment or support of public research institutions, etc.; or encourages enterprises to adopt specific technologies or high-tech through subsidies, funding coordination, tax and fee reductions, and other measures.
9. **Government procurement:** The government supports the development of domestic enterprises by prioritizing procurement of goods and services from the industry by the government or state-owned enterprises, procurement quotas, etc.
10. **Labor policy:** The government provides human resource assurance for industrial development through technical training programs, providing training subsidies, establishing or supporting training institutions, establishing skills committees, labor market access policies, migration policies and quotas, talent attraction supporting policies (such as priority household registration for talents, preferential housing for talents), wage tax credits, talent subsidies, etc.
11. **Infrastructure investment:** The government provides basic support for industrial development by constructing transportation infrastructure, information and communication technology infrastructure, energy infrastructure, and other areas.
12. **Promote industrial cluster:** The government promotes industrial agglomeration and development by constructing economic zones or industrial parks, supporting innovation clusters, industrial chain collaboration, etc. Additionally, the government encourages the development of key nodes in the industrial chain, provides upstream and downstream industry support, or simultaneously encourages the joint input and development of upstream and downstream industries to drive the entire industry.
13. **Environmental policy:** The government guides enterprises toward green development through environmental protection subsidies, energy conservation and emission

reduction policies, green credit, etc.

14. **Market access and regulation:** The government regulates market competition behavior by formulating product standards, industry access standards, implementing anti-monopoly policies, and maintaining fair competition policies.
15. **Consumer subsidy:** The government encourages industrial development through product subsidies, consumer stimulus subsidies, licensing support, and other measures. For example, but not limited to, encouraging trade-in of old products for new ones, promoting products in rural areas, issuing consumption vouchers, and providing supportive policies for electric vehicle licensing.
16. **Preferential land supply:** The government ensures land use needs for industrial development through preferential allocation of industrial land, land purchase discounts, rent subsidies, assisting enterprises in land consolidation, provision of utilities (such as water, power, roads), etc.
17. **Localization policy:** The government promotes local supply chains and economic development by requiring enterprises to employ local labor, operate locally, procure locally, etc.
18. **Promote entrepreneurship:** The government fosters emerging industries and innovative enterprises by building entrepreneurial service platforms such as incubators and accelerators, providing startups with facilities, funding, consulting, training, and other support.
19. **Industrial promotion:** The government provides display platforms for specific industries or enterprises by organizing or supporting trade exhibitions, product promotion events, etc., to expand product visibility and promote market development.
20. **Improving business environment:** The government reduces institutional transaction costs for enterprises by streamlining administration and delegating power, combining deregulation with enhanced services, optimizing approval processes; or by building a service-oriented government, enhancing the awareness and efficiency of government services for enterprises; or by improving the legal environment, increasing judicial efficiency, providing a fair business environment for enterprises, reducing transaction costs for enterprises; or by improving intellectual property protection mechanisms to enhance enterprises' motivation for R&D.

C.3 Conditionality

1. **Firm scale:** Policies specify the size of enterprises eligible for support, which can include conditions such as asset size, registered capital, number of employees, profitability, or tax contributions.

2. **Firm age:** Policies are designed to either support startups or target enterprises that have been operating for a specified number of years.
3. **R&D investment or specific technology:** Policies require that firms have a certain level of R&D investment or possess specific technological qualifications (e.g., patents, R&D platforms, or high-tech capabilities).
4. **Firm location:** Policies that are region-specific, meaning that only firms operating in a particular geographical area, paying taxes locally, or sourcing materials regionally are eligible.
5. **Firm ownership type:** Policies requiring certain types of ownership. For instance, some policies favor state-owned enterprises (SOEs), while others may prioritize domestic private firms, foreign-invested enterprises, or specific ownership structures.
6. **Specific firms:** Policies explicitly support a limited number of designated enterprises. These policies often name the supported firms directly.

C.4 Organizational Arrangements of Policy Implementation

C.4.1 Incentive Scheme

1. **KPI:** Using the assessment results of policy implementation as a crucial basis for evaluating lower-level governments, departments, and their leaders. These outcomes are critical criteria for officials' selection, promotion, and disciplinary actions. Notably, the assessment subjects focus on the performance of governments, departments, and officials, excluding enterprises or market entities.
2. **Supervision and inspection:** Conducting special supervision, regular inspection, organizing on-site examinations, and requiring lower-level governments and departments to regularly report work progress to higher-level departments during policy implementation.
3. **Positive incentives:** Commending and promoting exemplary models, providing financial rewards, giving preferential treatment in evaluations and selections, prioritizing support in project approvals, fund allocation, and linking with land quota allocations, etc.
4. **Negative incentives:** Specifying responsible units or individuals, issuing public criticisms to entities with unsatisfactory assessment results, establishing an accountability system, strictly holding accountable those who are ineffective in their work, severely punishing those with acts of dereliction of duty or misconduct, and implementing a "one-vote veto" system, etc.
5. **Setting target:** Setting clear and specific targets with indicators and numerical values, developing timeline, creating work logs, and breaking down and delegating tasks.

C.4.2 Experimentation and Learning

1. **Pilot & demonstration:** Selecting specific regions or units to carry out pilot projects, encourage pioneering and experimentation, supporting the demonstration bases, etc.
2. **Requiring local implementation:** Delegating approval authority to lower-level governments or relevant departments, granting decision-making discretion, encouraging them to formulate implementation plans independently, and encouraging their adjustments based on actual situations, etc.
3. **Encouraging innovation:** Encouraging lower-level governments and departments to engage in policy implementation innovation, allowing them to independently innovate in the process of policy execution, and promoting the development of innovative mechanisms and models.
4. **Allowing mistake:** During policy implementation, establishing fault tolerance mechanisms for lower-level governments and departments, exempting relevant responsibilities, showing leniency towards mistakes, and excluding such errors from negative evaluations or performance assessments.
5. **Learning experience:** Encouraging or requiring lower-level governments and departments to summarize their practices in policy implementation, promoting successful cases and innovative achievements, and forming replicable and scalable models, etc.

C.4.3 Local Condition

1. **Local adaptation:** Highlighting the importance of independently crafting policies tailored to the particular needs and circumstances of a locality, or devising support plans and detailed implementation strategies for higher-level government policies by aligning them with local conditions.
2. **Local input advantage:** Leveraging the region's advantages in factors such as geographical location, natural resources, labor force, capital, technology, business environment, and supply chain advantages to formulate policies and develop related industries.
3. **Local industry advantage:** Leveraging the existing local industrial base, prominent specialized sectors, key fields, and leading enterprises to promote related industries or establish regional industrial clusters.
4. **Differentiation:** Emphasizing the development of region-specific policies, offering targeted guidance, enacting enterprise-specific strategies, and devising tailored support policies based on the distinct circumstances of various industries.

C.4.4 Organizational Support

1. **Funding support:** Providing specific financial support for policy implementation, such as allocating special funds, increasing fiscal expenditure, broadening financing channels, securing matching funds, enhancing fund management, etc.
2. **Strict enforcement:** Highlighting the crucial need for policy execution, utilizing obligatory actions to mandate enforcement by subordinate government levels.
3. **Facilitating coordination:** Defining organizational arrangement (for example, forming specialized committees, drafting specific rules and processes, assigning departmental roles, proposing task requirements, highlighting focus areas, prioritizing efforts, planning work, assigning tasks, and scheduling workflows) along with establishing coordination and execution frameworks (such as integrated planning, creating interdepartmental coordination systems, regular meetings, joint inspections, information-sharing infrastructures, and collaborative problem-solving for crucial and complex issues).
4. **Institutional support:** Offering supportive policies by easing approval processes and enhancing key infrastructure; boosting capabilities through professional training and seminars for government staff to build a stronger talent pool; and advancing institutional development through system reforms and standardized management.

Appendix D Verification: Comparing with BERT and Random Forest

We use the human-labeled sample as the ground truth and benchmark our LLM against BERT models, a random forest model, and logistic regressions with different regularizations. We detail below how we implement the benchmark models and search for the best performance for each model.

Given the instability of the BERT model, we estimate and average the results from three runs and apply this procedure to all models for comparability. In each run, we use a consistent macro seed (1, 10, or 100) for sampling and algorithms, and we partition the human-labeled data into training (70%), validation (15%), and test (15%) sets. The split is stratified to maintain a constant proportion of positive and negative observations across all sets. We estimate model parameters on the training set, use the validation set for hyperparameter tuning and threshold selection, and reserve the test set for final performance evaluation.

D.1 BERT Model

We fine-tune hfl/chinese-bert-wwm-ext, a widely-used BERT model optimized for Chinese text. We add a classification layer to map the model's high-dimensional output to a binary prediction. We test two variants: a "fully-optimized" version where all model parameters are fine-tuned, and a "layer-optimized" version where only the new classification layer is trained.

Since documents often exceed the model's 512-token input limit, we use a sliding window approach. Documents are divided into overlapping 512-token segments. The window slides with a stride of 64 tokens; for example, after processing tokens 1–512, the next window processes tokens 65–576. This ensures significant overlap and allows context to be maintained across segments. The model generates a numerical representation (an embedding) for each segment, and these are averaged to create a single document-level vector.

Models are trained using the standard AdamW optimizer with a linear warmup and decay learning rate schedule. The learning rate follows a "linear warmup and decay" schedule, increasing for the first 6% of training steps before decaying, a practice that enhances training stability. To address the class imbalance between positive and negative examples in the training data, we use a WeightedRandomSampler that oversamples the minority class, helping to improve model performance. Hyperparameters (learning rate and weight decay) are tuned over 20 trials using the Optuna framework to maximize the performance measure of Area Under the Curve (AUC) on the validation set from the ranges of (1e-6, 1e-4) and (1e-6, 1e-2), respectively. The model outputs a continuous score between 0 and 1; we choose a decision threshold (e.g., classifying as positive if the score ¿ 0.6) to maximize the F1-score on the validation set.

D.2 Keyword-Based Models: Logistic Regression and Random Forest

We implement two standard classifiers using a Term Frequency-Inverse Document Frequency (TF-IDF) representation of the text. TF-IDF converts text into numerical vectors by weighting words based on their frequency within a document, discounted by their frequency across the entire corpus. We use the jieba library for Chinese word segmentation and include single words and adjacent word pairs (1- and 2-grams). We include terms that appear in at least one document (min_df=1) but in no more than 95 percent of documents (max_df=0.95). We apply sublinear term-frequency scaling to down-weight the influence of very high-frequency terms and use L2 normalization to ensure all document vectors have a unit norm, which helps mitigate the effect of outliers.

Logistic Regression We implement regularized logistic regression with three regularization penalty types: L1, L2, and Elastic Net, respectively. We use the SAGA solver,

which is efficient for sparse data. We use balanced class weights to improve performance. For each regularization type, we first search for optimal parameters by maximizing AUC via 5-fold cross-validation on the training set, and then choose the decision threshold for the binary prediction to maximize the F1-score on the validation data.

Random Forest The Random Forest model is an ensemble of decision trees. We tune a comprehensive set of hyperparameters using a grid search with stratified 5-fold cross-validation, optimizing for AUC. The grid includes parameters governing the number and structure of the decision trees: Number of trees in the forest (n_estimators): $\{300, 600, 900\}$, Maximum depth of each tree (max_depth): $\{\text{None}, 20, 40\}$, Minimum samples to split a node (min_samples_split): $\{2, 5\}$, Minimum samples required at a leaf node (min_samples_leaf): $\{1, 2\}$, Number of features to consider for a split (max_features): $\{\text{sqrt}, log2, 0.3\}$, Split quality criterion (criterion): $\{\text{gini}, \text{entropy}\}$. We choose the decision threshold for the binary prediction to maximize the F1-score on validation data.

From Table A1, our LLM approach achieves superior performance to all benchmark models across all evaluation metrics. Notably, it yields significant improvements in both precision and recall over the next-best alternative, BERT. The performance differences across model classes highlight the inherent complexity of classifying long, complex, unstructured policy text into industrial policy. In particular, the performance improves from the keyword approaches (Logistic regressions and Random Forest) to BERT that captures certain level of semantic complexity, and ultimately to the LLM approach.